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IoT-Enabled Solar-Powered Smart Wheelchair

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Certificate

It is certified that the project report has been prepared and written under my direct supervision and guidance. The project report is approved for submission for its evaluation.

Dr. Salim El Kkhediri

Dedication

This work is dedicated to everyone who has instructed us along our academic path, to everyone who has supported and walked beside us, to our beloved families for their endless help and patience. Additionally, we dedicate this effort to our supervisor.

Mansour Mutlaq Alharbi, Meshari Abdullah Alsaegh, and Saud Faisal Alfadda

Acknowledgement

First and foremost, all praises and thanks are due to Allah, the Almighty, for His blessings and for giving us the strength, patience, and ability to complete this project successfully.

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Mansour Mutlaq Alharbi, Meshari Abdullah Alsaegh, and Saud Faisal Alfadda

Abstract

This graduation project introduces the design and implementation of an **IoT-Enabled Solar-Powered Smart Wheelchair**, engineered to revolutionize mobility solutions for individuals with physical disabilities. By harnessing the power of **Internet of Things (IoT)** and **renewable solar energy**, the proposed system achieves sustainable, real-time health monitoring, intuitive control mechanisms, and enhanced safety measures—aligning seamlessly with **Saudi Vision 2030's** goals for sustainable development and smart city integration.

Key technological innovations include:

- **Real-Time Health Monitoring:** Advanced IoT sensors continuously track vital signs, enabling proactive health management and emergency response.
- **Gesture and Voice-Based Control:** Cutting-edge navigation systems allow for seamless control, enhancing user independence and reducing physical strain.
- **Obstacle Detection and GPS-Enabled Remote Tracking:** Intelligent multi-sensor systems provide precise navigation in complex environments, ensuring safety and reliability.
- **Solar-Powered Operation:** A fully integrated solar charging system promotes off-grid usability, reducing dependence on traditional power sources while minimizing carbon footprint.

A functional prototype is developed to validate the design's effectiveness in real-world scenarios, demonstrating **scalable deployment** in urban and rural areas. The solution is engineered with a strong emphasis on **cybersecurity**, **privacy protection**, and **reliability**, addressing critical concerns for connected medical devices.

This project is a step forward in redefining mobility aids, offering an **eco-friendly, cost-efficient, and intelligent** alternative that bridges gaps in traditional solutions. It positions itself as a key enabler for inclusive smart city development, expanding accessibility while contributing to the Kingdom's commitment to sustainable innovation. Furthermore, it sets the stage for future enhancements in autonomous mobility and smart healthcare systems, paving the way for smarter, greener urban living.

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Chapter 1

INTRODUCTION

INTRODUCTION

This graduation project presents the design and implementation of an **IoT-Enabled Solar-Powered Smart Wheelchair**. The primary objective is to enhance mobility support for individuals with physical disabilities, particularly in regions like Saudi Arabia, where there is a growing demand for advanced assistive technologies. Traditional wheelchairs face several limitations, including:

- Limited energy efficiency and dependence on conventional power sources.
- Lack of real-time health monitoring and user safety features.
- Incompatibility with the harsh environmental conditions of Saudi Arabia.
- Reduced independence and adaptability in off-grid areas.

To address these challenges, this project integrates **Internet of Things (IoT)** technology with **solar energy harvesting**, creating a sustainable and intelligent mobility solution. The main objectives of this integration are to:

- Enhance user autonomy and safety through real-time monitoring and obstacle detection.
- Improve energy efficiency by leveraging solar power for sustainable off-grid operation.
- Support Saudi Vision 2030 by contributing to sustainable development and technological innovation in assistive devices.

The proposed wheelchair utilizes solar panels to harness clean energy, which is stored and managed through an intelligent power distribution system. This eliminates the need for frequent grid-based charging, allowing for seamless operation in rural and remote areas. Additionally, IoT sensors provide real-time health monitoring and obstacle detection, enhancing user safety and independence.

This project aims to transform the concept of conventional wheelchairs by introducing smart, eco-friendly, and autonomous features, paving the way for a new era of sustainable assistive technologies in Saudi Arabia.

Globally, the increase in aging populations and rising rates of mobility impairments highlight the urgent need for innovative assistive technologies. According to the **World Health Organization (WHO)**,

by 2050, approximately 2.1 billion people will require mobility assistance, including over 900 million aged 60 and above [2]. This project aims to address these needs by providing a smart, sustainable, and energy-efficient solution that aligns with **Saudi Vision 2030**, contributing to improving the quality of life and promoting the use of clean energy technologies.

1.1 Problem Statement

The increasing global population of elderly individuals and people with physical disabilities has significantly amplified the demand for advanced assistive mobility technologies. However, conventional wheelchairs still rely on basic mobility features, lacking the technological advancements needed for modern living. These limitations are especially apparent in regions like Saudi Arabia, where environmental and infrastructural challenges are more severe. The key issues with traditional wheelchairs can be summarized as follows:

- **Lack of Real-Time Health Monitoring:** Traditional wheelchairs do not provide real-time tracking of vital health metrics, such as heart rate, oxygen levels, and body temperature, making it difficult to respond quickly in emergency situations.
- **Dependence on Electrical Charging:** Most wheelchairs require frequent charging from the electrical grid, which is not practical in rural or off-grid areas with limited infrastructure.
- **No Support for Sustainable Energy Solutions:** There is no integration of renewable energy like solar power, leading to high dependency on non-renewable energy sources. The proposed design will incorporate **Maximum Power Point Tracking (MPPT)** technology to optimize solar energy efficiency.
- **Limited Smart Connectivity and Monitoring:** Traditional designs do not support IoT-based monitoring or remote diagnostics, which limits caregiver intervention. The proposed wheelchair will utilize long-range communication technologies like **LoRaWAN** for better connectivity in remote areas.
- **Challenges in Harsh Environments:** Saudi Arabia's high temperatures and dusty conditions reduce the lifespan and performance of traditional wheelchairs. The new design will include IP-rated materials to protect against dust and extreme heat.

- **Difficult Navigation in Remote Areas:** Uneven terrain and lack of smart navigation features make it difficult for users to move independently in challenging environments. The proposed system will have enhanced obstacle detection and terrain adaptation.
- **Wireless Charging Capabilities:** To address the issue of charging in off-grid locations, the smart wheelchair will be equipped with **wireless solar charging modules** to ensure continuous operation.

In Saudi Arabia, these challenges are even more pronounced due to the vast rural areas and extreme climate. Many regions still lack reliable electricity, making electric wheelchairs impractical for everyday use. Furthermore, there is limited access to advanced healthcare facilities, which prevents real-time health monitoring for users who are often far from hospitals and clinics.

To address these issues, this project proposes the development of an **IoT-Enabled Solar-Powered Smart Wheelchair** that integrates solar energy for independent power, along with IoT technology for real-time health monitoring and smart navigation. This solution is designed to:

- Operate independently from electrical grids using solar power and MPPT technology.
- Provide continuous health monitoring with real-time alerts for heart rate, temperature, and oxygen levels.
- Enable smart navigation and obstacle detection in different terrains with high precision.
- Allow caregivers to monitor the wheelchair's status remotely via a mobile app, even in long-range settings.
- Support the goals of **Saudi Vision 2030** by promoting sustainable energy and improving the quality of life for individuals with disabilities.

This innovative design aims to enhance user independence, ensure safer mobility, and reduce the dependency on traditional energy sources, making it an ideal solution for the unique conditions in Saudi Arabia.

1.2 Challenges & Solutions

The successful development and deployment of an IoT-Enabled Solar-Powered Smart Wheelchair involve addressing several critical challenges across technical, economic, and environmental domains. These challenges must be effectively managed to ensure the project's success and sustainability, particularly within the context of Saudi Arabia. The main challenges and their proposed solutions are summarized in Table 1.1.

Challenge	Proposed Solution	Description
Solar Energy Efficiency	MPPT (Maximum Power Point Tracking)	Enhances energy harvesting from solar panels under varying sunlight conditions.
Stable IoT Connectivity	LoRaWAN and Mesh Networks	Ensures long-range communication in low-signal or remote areas.
Cybersecurity and Data Privacy	AES Encryption and TLS Protocols	Protects user data and ensures secure communication with remote servers.
Component Integration and Compatibility	EMI-Shielded Modular Design	Prevents interference and supports smooth integration of multiple IoT sensors.
Voice Recognition Accuracy	Noise-Cancellation and AI-based Filters	Improves speech recognition accuracy in noisy environments.
High Manufacturing Costs	Modular and Scalable Hardware	Reduces production costs and simplifies maintenance.
Maintenance and Operational Expenses	Use of Durable Materials and Self-Diagnostics	Minimizes repairs and ensures long-term reliability.
Harsh Weather Conditions	IP-Rated Enclosures and Dust Filters	Protects components from extreme heat, dust, and weather conditions.
Mechanical Durability	Lightweight Aluminum Alloy and Reinforced Structure	Ensures durability in rough terrains while maintaining a lightweight frame.

Table 1.1: Challenges and Proposed Solutions for the IoT-Enabled Solar-Powered Smart Wheelchair

Effectively addressing these challenges is crucial to delivering a reliable, affordable, and sustainable smart wheelchair solution that aligns with the environmental and societal needs of Saudi Arabia.

1.3 Advantages and Innovation

The IoT-Enabled Solar-Powered Smart Wheelchair introduces several innovative features and clear advantages, distinguishing it from existing mobility solutions. By integrating renewable solar energy with advanced IoT technologies, this project directly addresses mobility challenges for elderly and disabled individuals, particularly in Saudi Arabia, aligning closely with Vision 2030 objectives.

1. **Solar Energy Integration:** Utilizing Saudi Arabia's abundant sunlight [3], the wheelchair operates continuously without reliance on traditional electrical infrastructure. This sustainable energy solution ensures extended usage, even in off-grid and rural areas.
2. **Real-Time Health Monitoring:** Advanced IoT sensors are integrated to continuously track vital health parameters such as heart rate, blood pressure, and body temperature. Instant alerts are sent to caregivers and healthcare providers in case of emergencies, supporting proactive medical intervention.
3. **Intelligent Navigation System:** The wheelchair is equipped with ultrasonic sensors and GPS modules to detect obstacles, plan safe routes, and optimize navigation in various environments, including crowded urban areas and rough terrains.
4. **Adaptive Control Interfaces:** Designed with accessibility in mind, the wheelchair supports voice commands, gesture recognition, and mobile app control, allowing users to operate it seamlessly and intuitively based on their preferences and physical capabilities.
5. **Environmental Durability:** Built with weather-resistant materials and hybrid energy management, the wheelchair is designed to function reliably under extreme temperatures and dusty conditions typical in Saudi Arabia.
6. **Cybersecurity Measures:** To ensure the safety of sensitive medical data and user privacy, the system employs AES encryption and secure communication protocols to protect against unauthorized access and cyber threats.
7. **Healthcare System Integration:** The wheelchair is fully compatible with Electronic Health Records (EHR), allowing real-time health data to be synchronized with medical providers for seamless healthcare management.

8. **Modular and Cost-Effective Design:** The modular architecture allows for quick upgrades and easy maintenance, significantly reducing long-term costs while enhancing scalability for future enhancements.
9. **Lightweight and Compact Structure:** Designed using lightweight materials, the wheelchair maintains durability and ease of transportation, making it ideal for both indoor and outdoor mobility.
10. **Future-Proof Innovation:** Featuring a plug-and-play architecture, the system is designed to integrate emerging technologies easily, ensuring continuous improvement and alignment with future technological advancements.

Effectively integrating these advanced features not only improves the quality of life for users but also aligns with the sustainability goals of Saudi Vision 2030, supporting eco-friendly technologies and enhancing mobility solutions in the Kingdom.

1.4 Goals and Objectives

The primary goal of the IoT-Enabled Solar-Powered Smart Wheelchair project is to develop an advanced, sustainable, and intelligent mobility solution tailored specifically for elderly and physically disabled individuals, particularly within the Saudi Arabian context. This initiative aligns closely with Saudi Vision 2030, emphasizing sustainability, inclusivity, and technological innovation.

1.4.1 Main Goals

- **Sustainability through Renewable Energy:** Maximize the use of solar energy to enable off-grid operation and reduce dependency on traditional power sources.
- **Intelligent Monitoring and Navigation:** Integrate IoT for real-time health monitoring and GPS-based smart navigation.
- **Durability in Harsh Conditions:** Design the wheelchair to withstand Saudi Arabia's extreme temperatures and dusty environments.

- **Cost-Effective and Modular Design:** Ensure low maintenance costs and easy component replacement through modular architecture.

1.4.2 Specific Objectives

Objective	Target Metric
Achieve 30% higher energy efficiency	Compared to traditional electric wheelchairs through optimized solar utilization.
Ensure 95% uptime in IoT connectivity	For real-time monitoring and diagnostics, even in remote areas.
Implement an 80% Increase user independence by 90%	Through adaptive control interfaces like voice and gesture recognition.
Support off-grid operation for 10 hours	Operate reliably in rural areas on a single solar charge.
Reduce operational costs by 25%	Through modular design and renewable energy integration.
Enable 85% accuracy in health monitoring	Real-time data analysis for personalized healthcare and maintenance.
Reduce dependency on non-renewable energy by 50%	Supporting Vision 2030's sustainability goals.
Introduce at least one patentable innovation	Through AI-driven navigation and modular upgrades.

Table 1.2: Specific Objectives for the IoT-Enabled Solar-Powered Smart Wheelchair

1.5 Study Scope

This study focuses on the theoretical design and simulation of the **IoT-Enabled Solar-Powered Smart Wheelchair**, with a clear emphasis on enhancing mobility for individuals with disabilities and the elderly, particularly in Saudi Arabia. The scope of the study includes the following key areas:

- **Integration of IoT Technologies:** Implementing real-time health monitoring, remote control, and cloud-based data synchronization for enhanced mobility and safety.
- **Deployment of Solar Panels:** Utilizing solar energy to enable off-grid operation, reducing dependence on traditional power sources and promoting sustainability.
- **Development of Navigation Systems:** Designing smart navigation with obstacle detection, path planning, and real-time route optimization for improved user safety.
- **Assessment of System Reliability:** Testing the wheelchair's performance under simulated Saudi Arabian environments, including heat, dust, and rough terrains.

- **Preparation for Practical Implementation:** Laying the groundwork for prototype development and field testing in the next semester.

The study aligns with the goals of Saudi Vision 2030, focusing on sustainability, inclusivity, and advanced mobility solutions.

1.6 Study Plan

In this section, we present the project plan for Phase 1.

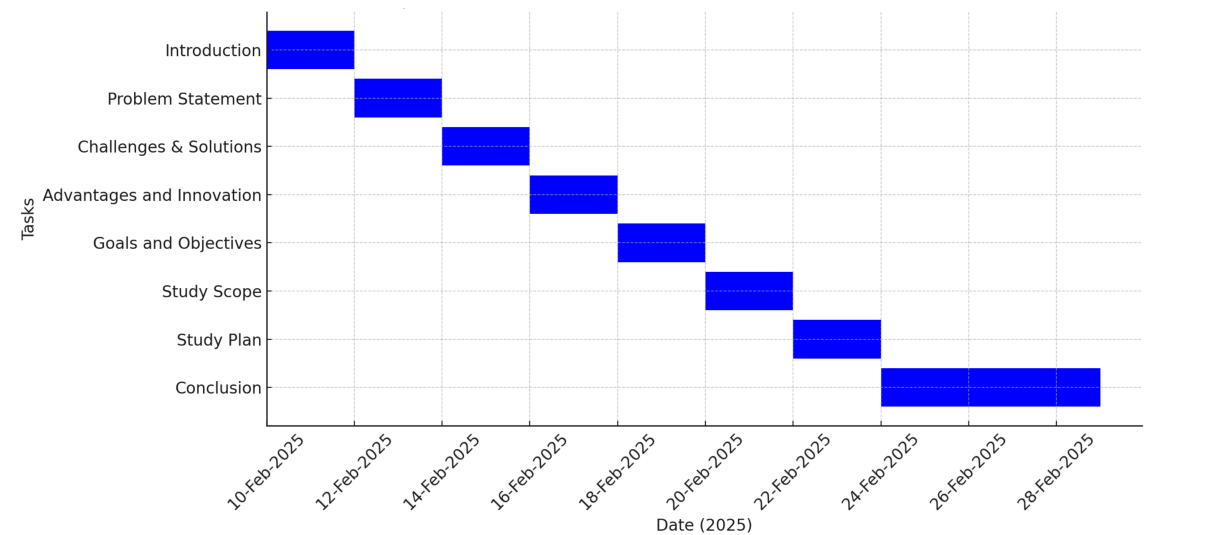


Figure 1.1: Chapter 1

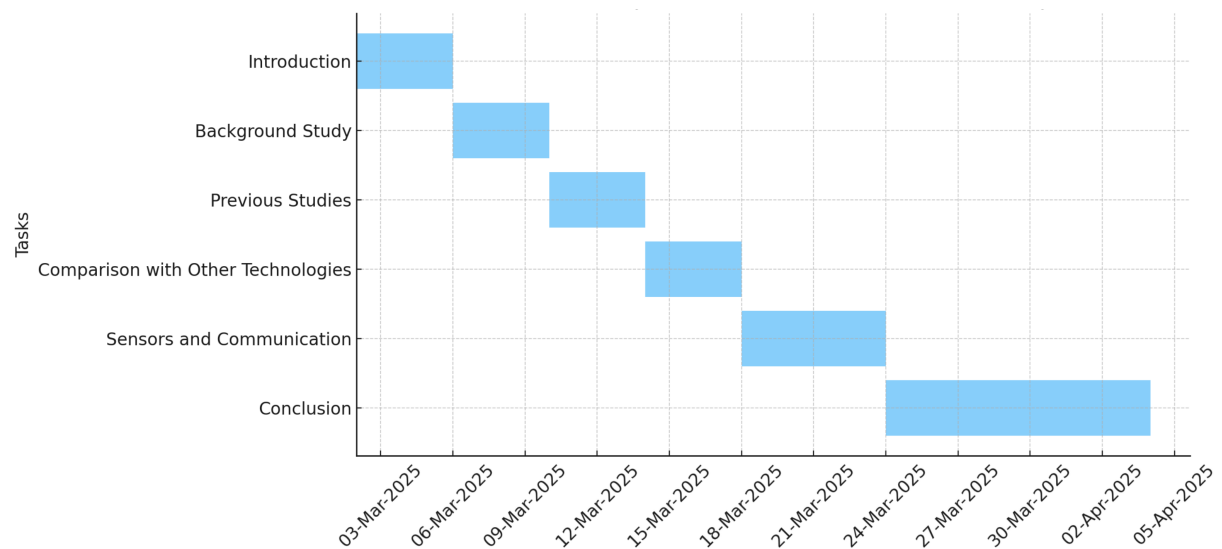


Figure 1.2: Chapter 2

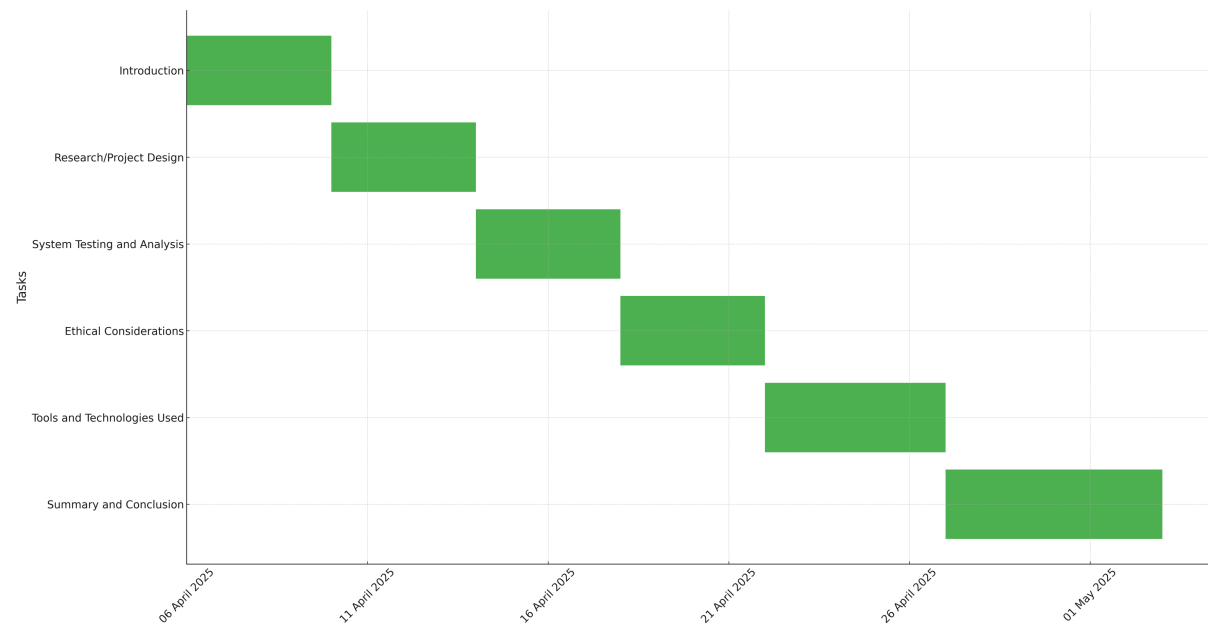


Figure 1.3: Chapter 3

1.7 Conclusion

This chapter presented a comprehensive overview of the **IoT-Enabled Solar-Powered Smart Wheelchair**, highlighting its potential to transform mobility for individuals with disabilities and the elderly in Saudi Arabia. The study scope outlined the primary technological integrations, including IoT, solar energy, and smart navigation, while the study plan detailed the project's phased approach from theoretical design to practical implementation. This initiative not only supports the objectives of Saudi Vision 2030 but also promises a sustainable, intelligent, and cost-effective solution for mobility assistance. The next chapter will explore the **Literature Review**, providing insights into existing technologies and innovations in smart wheelchair design.

Chapter 2

LITERATURE REVIEW

LITERATURE REVIEW

2.1 Introduction

This chapter provides a critical exploration of existing literature on IoT-based assistive technologies and solar-powered mobility solutions, with a particular focus on applications relevant to the IoT-Enabled Solar-Powered Smart Wheelchair. The purpose of this review is to evaluate current technological advancements, identify research gaps, and assess the practical implications of integrating IoT with renewable energy in the context of Saudi Arabia's growing demand for sustainable assistive devices.

The literature review is structured as follows:

- **Background:** A foundational understanding of IoT applications, solar energy utilization, and their role in enhancing assistive mobility.
- **Previous Studies:** An in-depth analysis of smart wheelchair technologies, identifying strengths, limitations, and areas for future innovation.
- **Comparison with Existing Technologies:** A comparative analysis of different mobility solutions, emphasizing the unique contributions of the proposed project.

Through this structured analysis, the chapter aims to demonstrate the critical need for innovative solutions like the IoT-Enabled Solar-Powered Smart Wheelchair, highlighting its alignment with Saudi Vision 2030 and global sustainability initiatives.

2.2 Background

The Internet of Things (IoT) is a technological paradigm that connects physical devices to the internet, enabling them to collect, share, and act on data automatically through embedded sensors and wireless communication. In this project, titled "*IoT-Enabled Solar-Powered Smart Wheelchair*", IoT technology is leveraged to enhance traditional wheelchairs with smart, autonomous capabilities. The main features include:

- **Health Monitoring:** Real-time tracking of vital signs, enabling immediate health data collection and proactive response.

- **Obstacle Detection:** Integration of sensors for real-time detection and avoidance of obstacles during navigation, ensuring safer mobility.
- **Semi-Autonomous Navigation:** IoT-based path planning and remote control to facilitate smoother and more efficient movement.

To ensure sustainable and uninterrupted operation, solar power is integrated, especially suitable for regions like Saudi Arabia, where sunlight exceeds 3,000 hours annually. This renewable energy source allows for off-grid capabilities, reducing reliance on traditional power sources and supporting Vision 2030's focus on sustainable technology.

System Architecture Overview: Figure 2.1 represents the block diagram of the IoT-Enabled Solar-Powered Smart Wheelchair, illustrating the flow of power and data between components. The system comprises:

- **Solar Panels:** Harvest sunlight and convert it into electrical energy.
- **Charge Controller:** Regulates the power flow to prevent overcharging.
- **Battery Storage:** Stores the harvested energy for consistent operation.
- **Power Management Unit:** Manages energy distribution across all components.
- **Microcontroller:** Acts as the central processing unit, coordinating sensor data, IoT modules, and user commands.
- **Sensors and Actuators:** Include health monitoring, obstacle detection, and mobility control.
- **Data Communication Module:** Facilitates real-time data exchange with cloud-based monitoring systems.

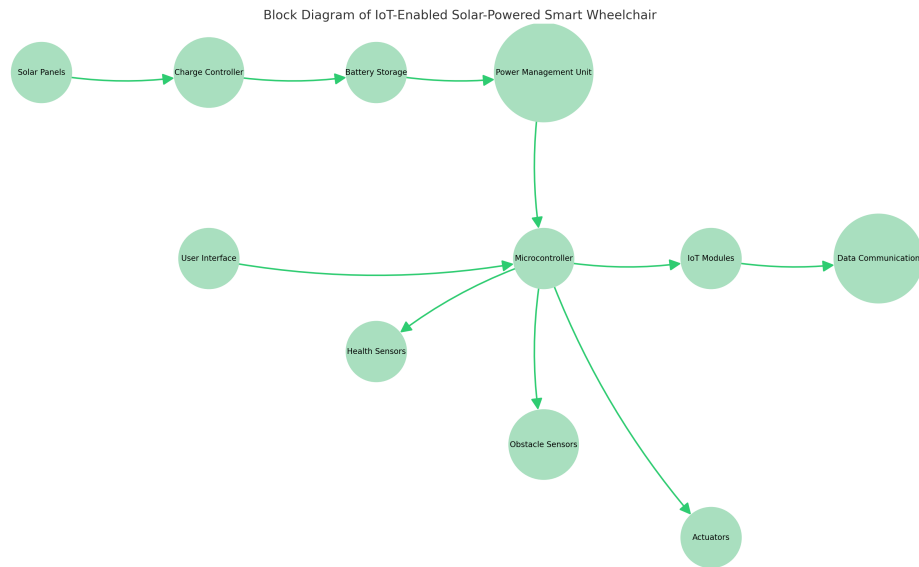


Figure 2.1: Block Diagram of IoT-Enabled Solar-Powered Smart Wheelchair

Recent advancements have demonstrated the feasibility of integrating solar energy into smart wheelchairs. A study by M. Pita outlines the design and manufacturing process of a solar-powered wheelchair, emphasizing energy efficiency, extended battery life, and off-grid capabilities [4]. The research highlights the use of photovoltaic (PV) panels to harness solar energy, which is stored in high-capacity batteries to ensure continuous operation during low sunlight conditions. Efficient power management systems optimize energy consumption and extend the wheelchair's operational range.

However, certain challenges must be addressed to ensure reliability and effectiveness:

- **Energy Storage Reliability:** Maintaining consistent power during cloudy or low-sunlight periods.
- **Data Security and Transmission:** Protecting sensitive health and location data from cyber threats.
- **Environmental Durability:** Ensuring robust operation in extreme weather conditions typical of Saudi Arabia.

Addressing these challenges through durable hardware, smart energy management, and secure communication protocols is crucial for reliable and sustainable wheelchair operation.

2.2.1 Internet of Things (IoT) in Assistive Technology

The Internet of Things (IoT) plays a foundational role in enhancing assistive technologies by integrating smart sensors and communication modules into daily-use devices. For the *IoT-Enabled Solar-Powered Smart Wheelchair*, IoT provides intelligent functionalities that significantly improve user experience, autonomy, and safety.

Key Features of IoT in Smart Wheelchairs:

- **Real-Time Health Monitoring:** Continuous tracking of vital signs such as heart rate, body temperature, and blood pressure. This data is transmitted to caregivers or cloud-based applications for monitoring and emergency response.
- **Obstacle Detection:** Equipped with ultrasonic or LiDAR sensors to identify obstacles in real-time, enhancing navigation safety in both indoor and outdoor environments.
- **Location-Aware Navigation:** GPS modules enable precise location tracking, which is crucial for route planning and emergency location sharing.
- **Remote Control and Alerts:** Mobile and web-based dashboards allow caregivers to monitor wheelchair status, control movement remotely, and receive real-time alerts for critical events.



Figure 2.2: IoT-based assistive technologies for people with disabilities, showcasing real-world use cases like smart glasses, connected canes, wheelchairs, ramps, and bus stop systems [1].

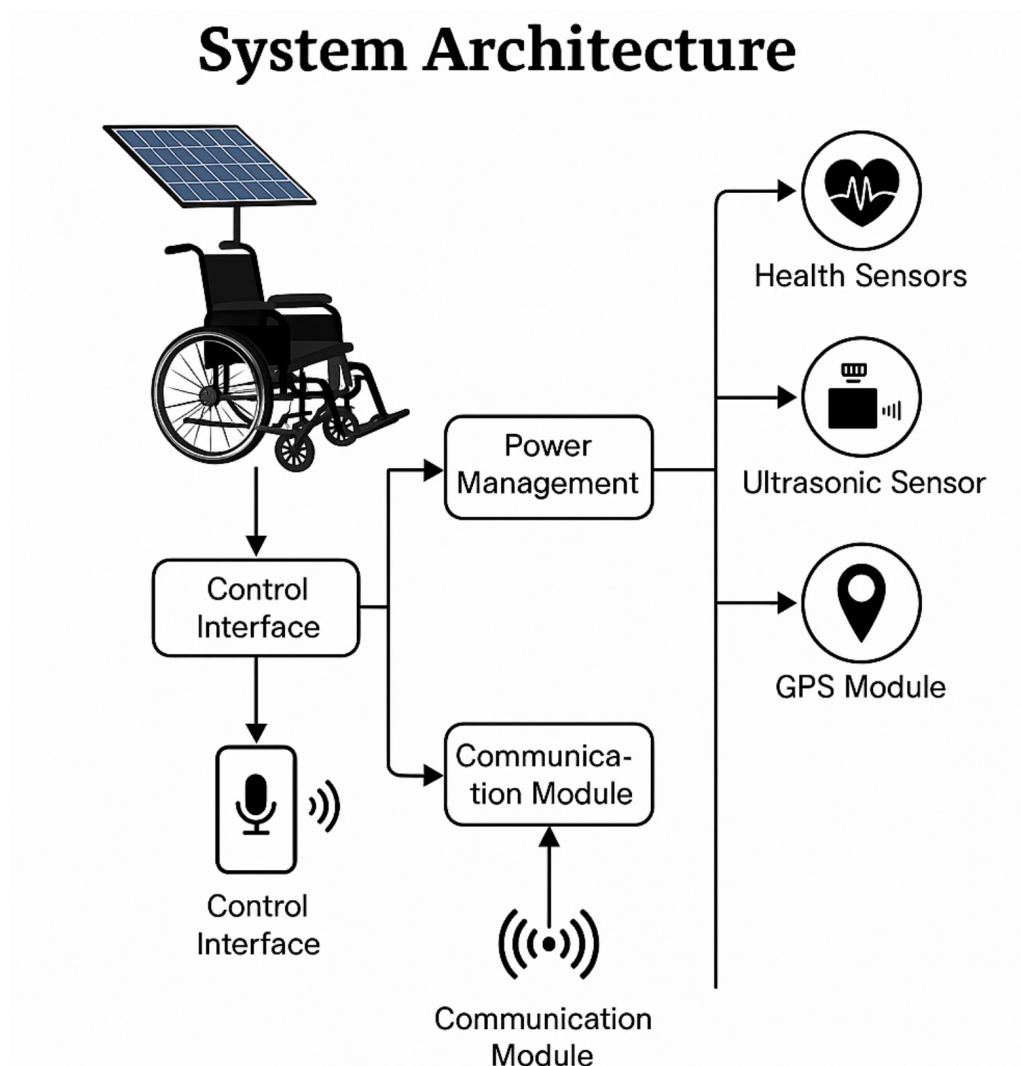


Figure 2.3: System Architecture of the IoT-Enabled Solar-Powered Smart Wheelchair. The diagram shows the integration of the photovoltaic panels, power management, control interface, communication module, health sensors, ultrasonic sensors, and GPS module for efficient and autonomous operation.

2.2.2 Data Communication in IoT-Enabled Systems

Effective data communication is crucial in IoT-enabled smart wheelchairs to ensure real-time monitoring, control, and feedback mechanisms. The system utilizes various modules such as GPS for location tracking, IoT sensors for health and environmental monitoring, and cloud servers for data processing and storage. Figure 2.4 illustrates the complete data communication flow.

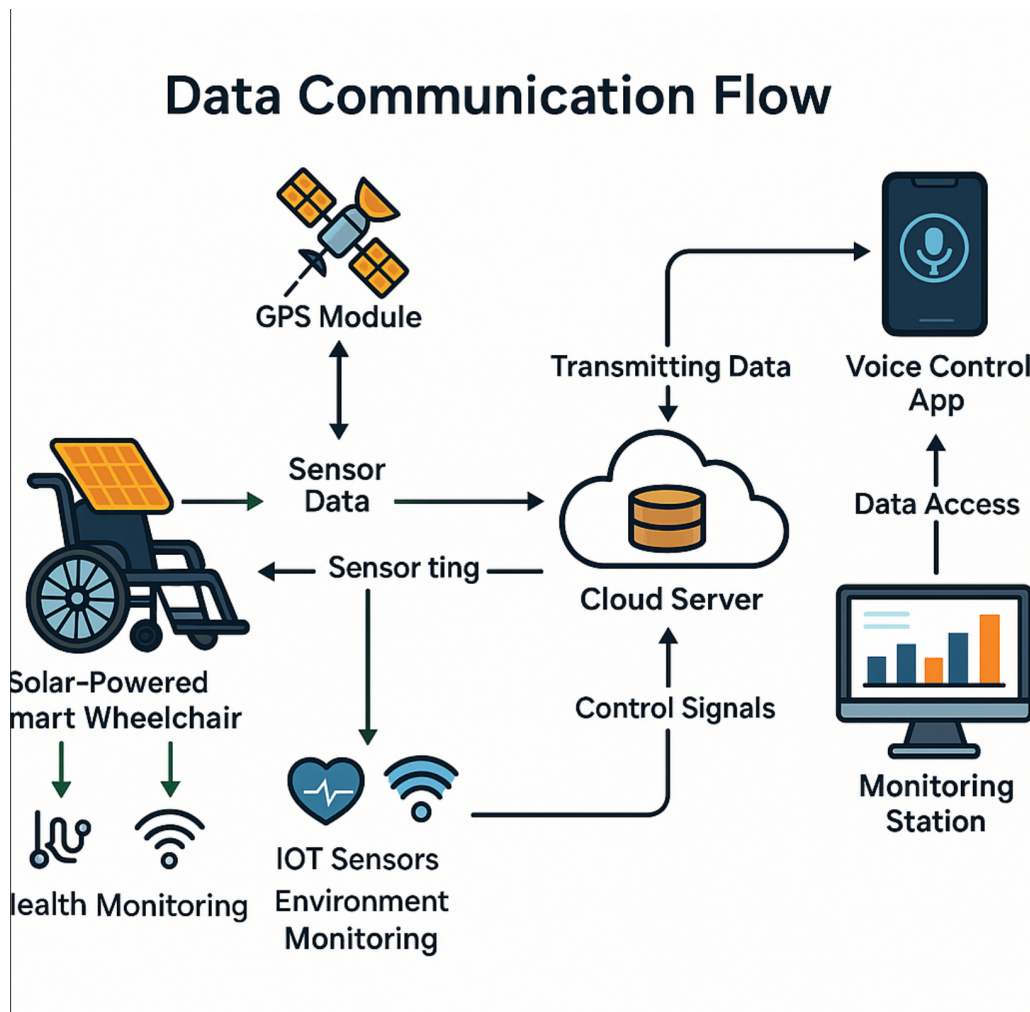


Figure 2.4: Data Communication Flow in the IoT-Enabled Solar-Powered Smart Wheelchair. Sensor data from the wheelchair is transmitted to the cloud for processing and monitoring, enabling real-time health tracking, environmental sensing, and navigation.

Related Work: Bhatnagar et al. (2022) developed a solar-powered wheelchair equipped with IoT modules, demonstrating enhanced user mobility and energy efficiency [5]. Their study showed how integrating IoT with renewable energy can significantly extend the operational range of mobility devices, especially in off-grid scenarios.

Challenges in IoT Integration:

- **Data Privacy and Security:** Ensuring the confidentiality of health metrics during wireless transmission.
- **Network Stability:** Maintaining continuous operation during network interruptions.
- **Low Latency for Real-Time Response:** Minimizing delays in sending alerts and executing commands.

mands, which is critical for safety.

Proposed Solutions:

1. Lightweight Communication Protocols: Technologies like MQTT (Message Queuing Telemetry Transport) and BLE (Bluetooth Low Energy) are crucial in IoT applications for efficient and low-power communication.

- **MQTT:** Uses a publish/subscribe model, reducing data transfer overhead and ensuring efficient communication between devices.
- **BLE:** Suitable for short-range, low-power communication, making it ideal for wearable and portable IoT devices.

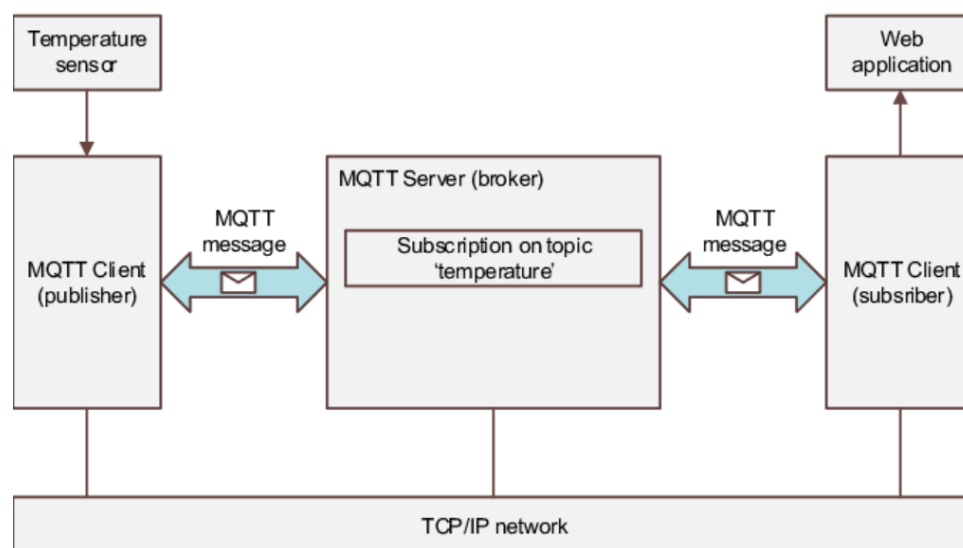


Figure 2.5: MQTT Protocol Communication Model for IoT Systems. (Image sourced from ResearchGate as a demo illustration.)

2. Encrypted Data Transmission: Ensuring data security is essential in IoT systems, especially when handling sensitive information.

- **AES (Advanced Encryption Standard):** Provides strong symmetric encryption to protect data integrity.
- **TLS/SSL (Transport Layer Security):** Secures data transmission over networks, preventing unauthorized access.

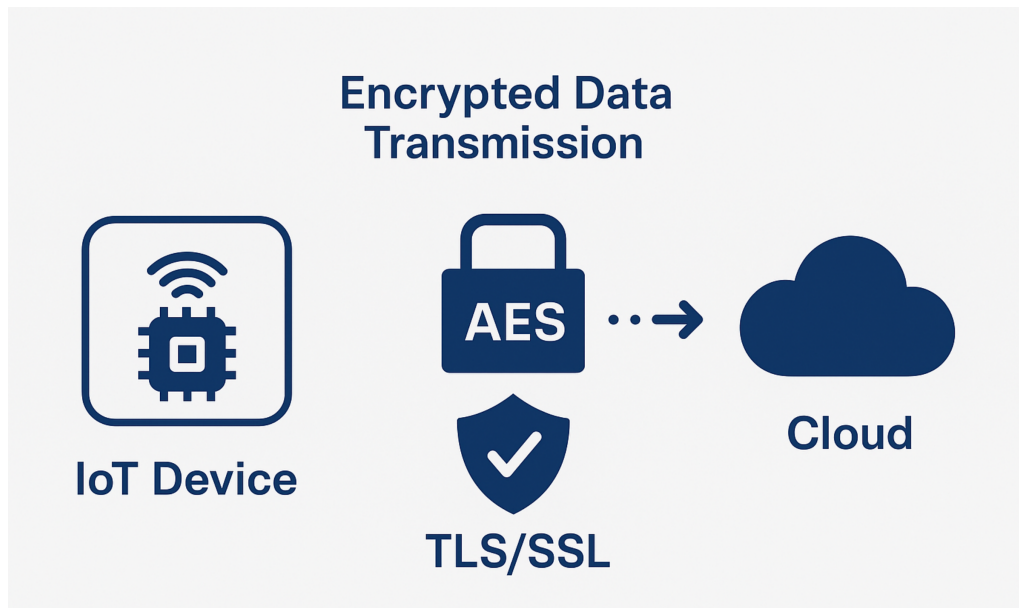


Figure 2.6: Encrypted Data Transmission in IoT Systems using AES and TLS/SSL. This diagram illustrates secure communication from IoT devices to cloud infrastructure.

3. Local Processing for Critical Functions: To maintain reliable operation even during network disruptions, microcontrollers handle critical functions locally.

- **Real-Time Processing:** Immediate response to critical tasks without relying on cloud services.
- **Reduced Latency:** Local processing minimizes delay, enhancing system responsiveness.
- **Data Integrity:** Maintains operational continuity when network connectivity is lost.

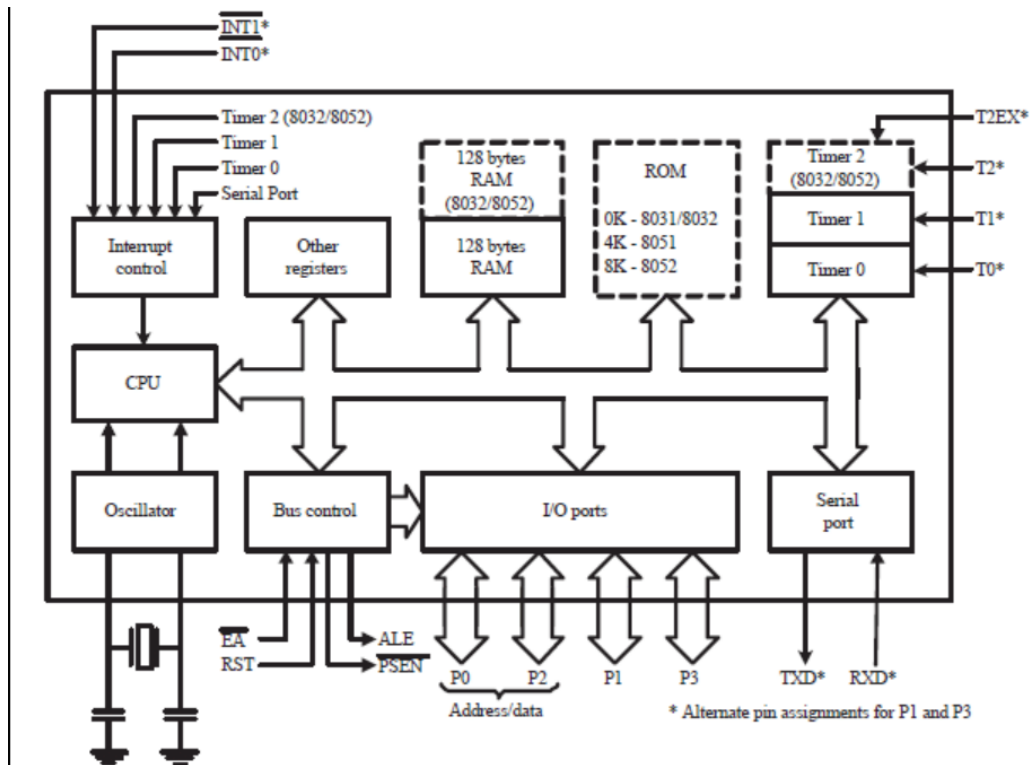


Figure 2.7: Block Diagram of the Intel 8051 Microcontroller Architecture. (Image sourced from ResearchGate as a demo illustration.)

IoT integration is not merely a luxury; it is a fundamental requirement to transform wheelchairs from basic mobility aids into intelligent, adaptive, and responsive devices that cater to both user needs and environmental challenges.

2.2.3 Solar Power Integration in Mobility Aids

The integration of solar power provides a sustainable and cost-effective solution for extending wheelchair operation without dependency on conventional grid-based charging. This approach is particularly beneficial in regions with abundant sunlight, such as Saudi Arabia.

Why Solar Power in This Project?

- **High Sunlight Availability:** Saudi Arabia receives over 3,000 hours of sunlight annually, making it an ideal candidate for solar energy utilization.
- **Off-Grid Power Solutions:** Solar energy allows operation in rural or underdeveloped areas with limited grid infrastructure.

- **Alignment with Vision 2030:** Supports green energy initiatives under Saudi Vision 2030, promoting sustainability and energy independence.

System Components Powered by Solar Energy:

- Motors for smooth and efficient mobility.
- Sensors for obstacle detection and health monitoring.
- IoT communication modules for real-time data exchange.

Supporting Study: Bhatnagar et al. (2022) demonstrated the effectiveness of solar energy in assistive devices, showing that flexible solar panels significantly extend operational time while reducing dependency on frequent charging [5].

Challenges of Solar Integration:

- **Reduced Efficiency:** Performance drops in cloudy or dusty environments.
- **Weight Considerations:** Added weight from photovoltaic panels.
- **Thermal Stress:** High temperatures may affect battery efficiency and panel lifespan.

Proposed Solutions:

- **Lightweight High-Efficiency Panels:** Reduces load while maximizing energy capture.
- **MPPT (Maximum Power Point Tracking):** Optimizes energy harvesting during variable sunlight conditions.
- **Hybrid Storage Systems:** Combines solar power with batteries and smart power management for consistent operation.

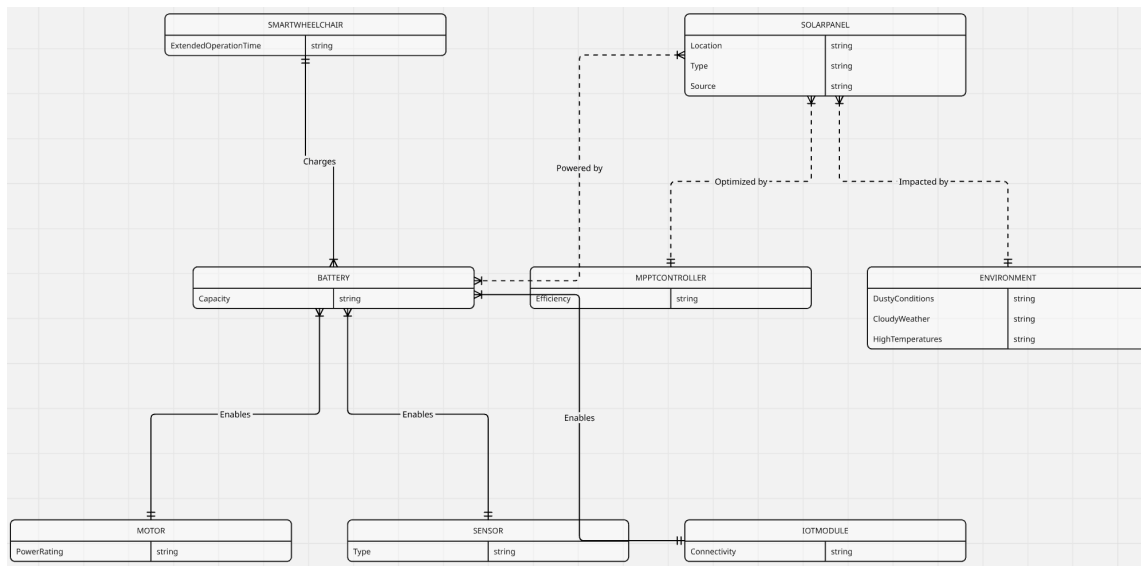


Figure 2.8: ER Diagram for Solar Power Integration in Smart Mobility Aids. It illustrates how solar energy is harvested, optimized, and stored to power key wheelchair components such as motors, sensors, and IoT modules, while accounting for environmental factors.

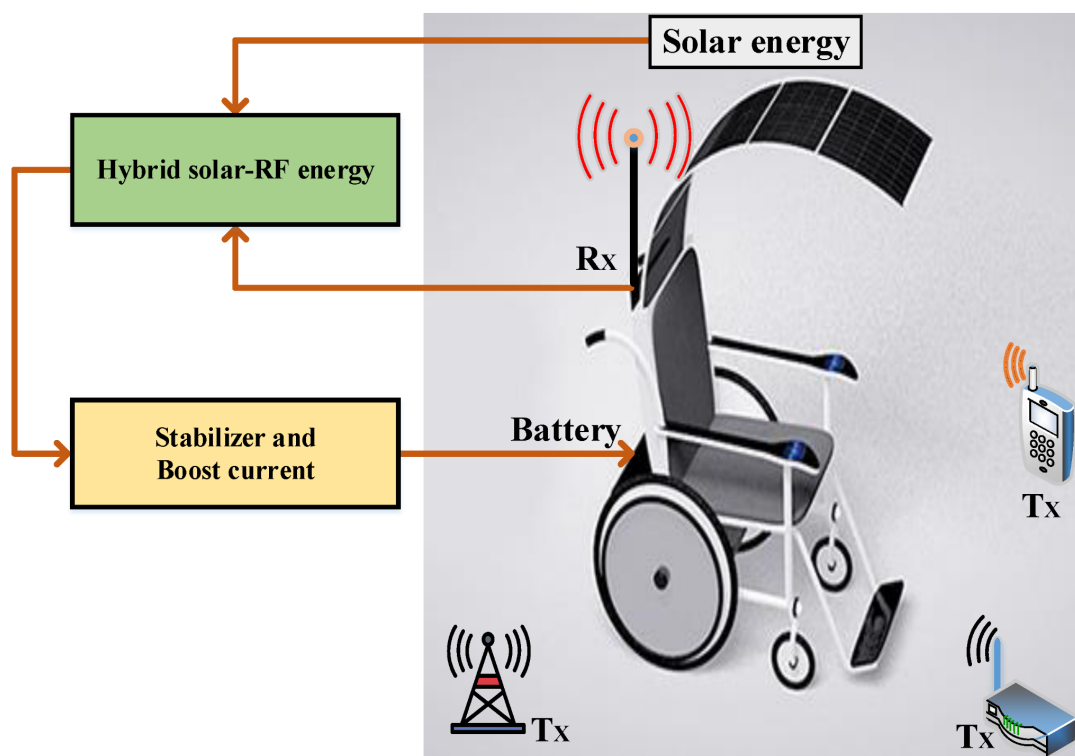


Figure 2.9: Hybrid Solar-RF Powered Wheelchair System, showing energy harvesting, regulation, and storage for wheelchair operation. Source: Electronics (MDPI, 2021), used for academic purposes.

2.2.4 Energy Management and Solar Power Integration

Solar energy is a sustainable and practical power source for smart wheelchairs, especially in regions with high solar exposure like Saudi Arabia. The integration of photovoltaic (PV) panels allows the wheelchair to harvest solar energy, converting it into electrical power to drive the system efficiently. Figure 2.10 illustrates the energy flow within the system.

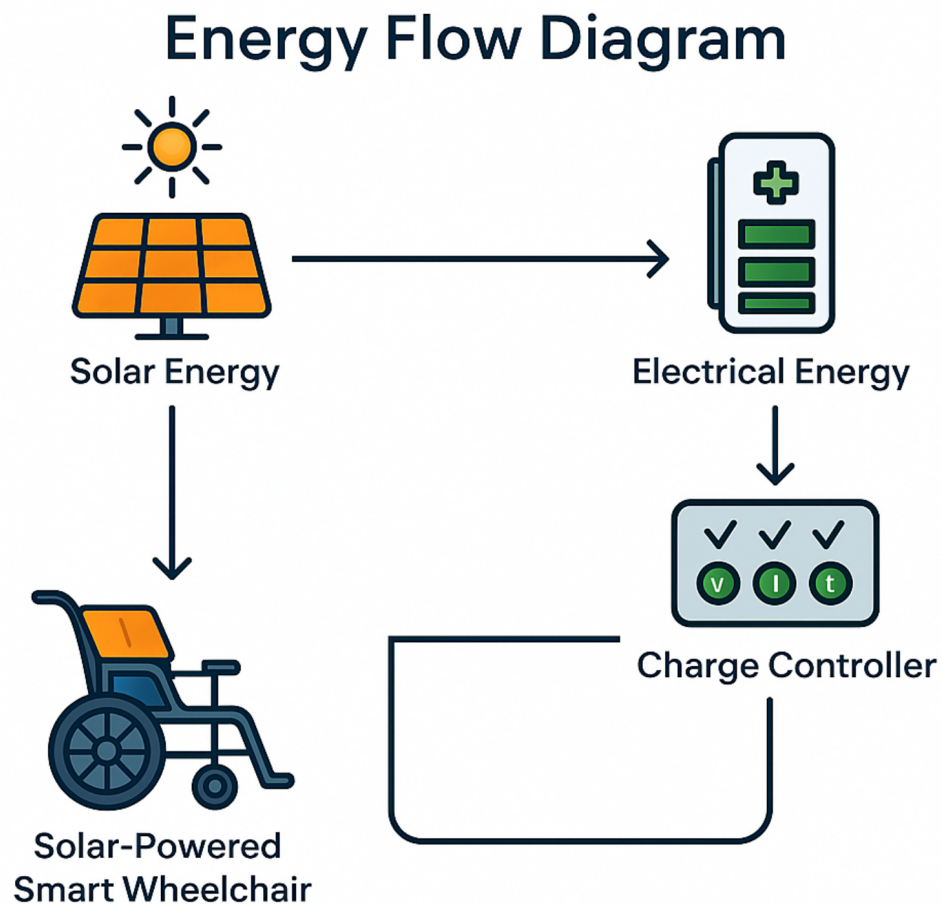


Figure 2.10: Energy Flow Diagram for the IoT-Enabled Solar-Powered Smart Wheelchair. Solar energy is captured by the photovoltaic panels, converted into electrical energy, regulated by a charge controller, and finally distributed to power the wheelchair.

2.2.4.1 Assistive Technology Needs in Saudi Arabia

Saudi Arabia has a growing demand for assistive technologies that are:

- Technologically advanced and adaptive to local conditions.

- Designed to withstand harsh environmental factors such as extreme heat and dust.
- Capable of operating independently from unstable or limited grid power sources.

According to the **General Authority for Statistics**, more than 7% of the Saudi population has some form of disability. Despite this demand, access to assistive technologies remains limited, particularly in rural or underserved areas where infrastructure is weak and imported solutions are often impractical.

Limitations in the Current Market:

- **Climate Unsuitability:** Imported wheelchairs are not optimized for extreme temperatures and dusty environments.
- **Frequent Charging Requirements:** Most models require continuous grid-based recharging, which is impractical for off-grid locations.
- **Limited Customization and Support:** Imported solutions often lack local technical support and are costly to maintain.

How Our Project Solves These Gaps: The **IoT-Enabled Solar-Powered Smart Wheelchair** is specifically designed to address these challenges:

- **IoT Integration:**
 - Real-time health monitoring for user safety.
 - GPS-based navigation to enhance mobility and location awareness.
 - Emergency safety alerts for rapid response.
- **Solar-Powered Operation:** Reliable off-grid power through solar energy, ensuring continuous use in rural areas.
- **Vision 2030 Alignment:** Promotes inclusivity for individuals with disabilities and supports environmental sustainability.

Supporting Research: Roy (2024) demonstrated the viability of low-cost IoT systems for health monitoring in mobility-impaired patients[6]. Our project extends this concept by integrating solar power to achieve autonomous, off-grid functionality, particularly suited for the Saudi Arabian landscape.

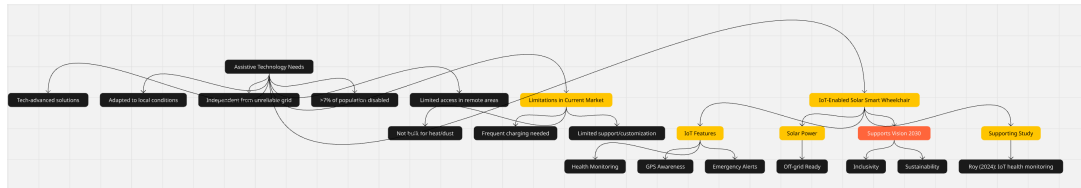


Figure 2.11: Block diagram summarizing assistive technology needs, market limitations, and smart wheelchair solutions.



Figure 2.12: Illustration of a solar-powered IoT-enabled smart wheelchair adapted for Saudi Arabian conditions.

2.3 Previous Studies

2.3.1 Related Work on Smart Wheelchairs

The integration of **Internet of Things (IoT)** and intelligent sensors in smart wheelchairs has demonstrated significant improvements in mobility and independence for users with disabilities. Recent studies have explored the design and manufacture of **solar-powered smart wheelchairs**, integrating advanced IoT modules to enhance functionality and safety. Key components of these systems include:

- **Microcontrollers:** Efficiently manage sensor data processing and communication tasks.
- **NodeMCU Modules:** Enable seamless cloud-based communication for real-time data access.
- **GPS Modules:** Provide real-time location tracking and optimized navigation paths.
- **Ultrasonic Sensors:** Detect obstacles and ensure safe mobility across various terrains.

The study emphasizes how these components collectively transform traditional wheelchairs into intelligent, responsive devices capable of **autonomous navigation** and **real-time health monitoring**. Furthermore, the integration of **Photovoltaic (PV) panels** plays a critical role by providing sustainable energy to power the wheelchair independently from electrical grids. This design is particularly advantageous in regions with abundant sunlight, such as **Saudi Arabia**, where solar energy presents a highly viable solution for extending wheelchair operation time and minimizing reliance on frequent charging.

This work aligns directly with the objectives of the *IoT-Enabled Solar-Powered Smart Wheelchair* project, which emphasizes energy efficiency, real-time monitoring, and autonomous navigation as core features to support individuals with disabilities in diverse and challenging environments.

noindent• **Control Mechanisms and User Interfaces:** The design of accessible and adaptive control systems is crucial for smart wheelchairs, especially for users with severe mobility impairments. One notable study introduced a wheelchair control system based on **eye gaze technology**, allowing users to operate the wheelchair through eye movements rather than traditional joysticks. This approach significantly improves accessibility and autonomy for individuals with limited physical capabilities. Additionally, **voice commands** and **gesture-based navigation** have been implemented to simplify control mechanisms, making smart wheelchairs more intuitive and user-friendly.

- **Energy Efficiency and Solar Power Solutions:** Sustainable energy solutions are essential for the long-term operation of smart wheelchairs, particularly in regions with abundant sunlight such as Saudi Arabia. Research has demonstrated the integration of **retractable solar panels** into electric wheelchairs, utilizing photovoltaic cells to harness solar energy and power DC motors and batteries. This technology reduces the need for frequent charging and promotes off-grid operation, aligning well with the sustainability goals of Saudi Vision 2030.
- **Safety and Obstacle Detection Systems:** Ensuring user safety through advanced obstacle detection is a critical aspect of smart wheelchair design. IoT-enabled wheelchairs have incorporated **ultrasonic sensors** and **GPS modules** to detect obstacles in real-time, facilitating safe navigation across various terrains. These systems help users avoid collisions and navigate crowded or uneven spaces effectively, enhancing overall user safety and confidence.
- **Health Monitoring and IoT Connectivity:** The integration of health monitoring systems with IoT connectivity enables continuous tracking of user well-being and remote support. Modern smart wheelchairs are equipped with **pulse sensors**, **LDR sensors**, and **ultrasonic modules** for real-time health monitoring and environmental awareness. IoT platforms are used to transmit this data to caregivers or medical professionals, ensuring rapid response during emergencies and promoting proactive health management.
- **Local Context: Assistive Technology Needs in Saudi Arabia:** The demand for advanced assistive technologies in Saudi Arabia has grown significantly, driven by demographic changes and national initiatives like **Vision 2030**. Despite global advancements, many imported smart wheelchairs are not fully adapted to the local environment, particularly in terms of climate, infrastructure, and cultural needs. Saudi Arabia's harsh climatic conditions, such as high temperatures and dusty environments, necessitate robust and efficient mobility solutions. There is also an increasing interest in leveraging IoT and solar energy to enhance the sustainability and autonomy of assistive devices tailored to the Kingdom's environmental challenges.
- **Synthesis and Research Gaps:** The reviewed literature highlights significant progress in smart wheelchair technologies, including IoT integration, gesture-based control, multi-mode sensing, and solar-powered operation. However, critical research gaps remain, particularly in:
 - **Cost and Complexity:** Many existing solutions are too costly or technologically complex for adoption in developing regions, including Saudi Arabia.

- **Environmental Adaptability:** Limited research on the long-term reliability of smart wheelchairs in extreme environments like high temperatures and dust storms.
- **User-Centered Design:** Lack of direct feedback from local users during the development process.
- **Renewable Energy Optimization:** More investigation is needed to optimize the integration of solar panels with IoT for maximum efficiency and cost-effectiveness.

2.4 Comparison with Other Technologies

To clearly illustrate the strengths of the proposed smart wheelchair design, a comparative analysis was conducted between four major wheelchair types:

- **Traditional Electric Wheelchairs**
- **IoT-Enabled Wheelchairs**
- **Solar-Powered Wheelchairs**
- **Proposed IoT + Solar-Powered Smart Wheelchair**

The comparison is based on the following key aspects:

- **Power Source:** Whether the wheelchair relies on grid electricity, solar energy, or hybrid sources.
- **User Monitoring Capabilities:** Ability to monitor user's health status in real-time.
- **Navigation Methods:** Manual, joystick-based, or GPS/IoT-assisted navigation.
- **Obstacle Detection Systems:** Availability of smart sensors for collision avoidance.
- **Energy Efficiency:** Efficiency in power consumption during operation.
- **Environmental Sustainability:** Degree of support for green and renewable energy.
- **Suitability for Saudi Arabia:** How well each technology adapts to the region's climate, infrastructure, and societal needs.

This structured comparison highlights the innovation and practical benefits of the proposed design, particularly in solving:

- Environmental challenges (e.g., high temperatures and dusty conditions)
- Infrastructural limitations in remote or off-grid areas
- Health monitoring and safety for users with mobility impairments

Table 2.1: Comparison of smart wheelchair technologies and features

Feature	Traditional Electric Wheelchair	IoT-Enabled Wheelchair	Solar-Powered Wheelchair	IoT + Solar-Powered Smart Wheelchair (Proposed)
Power Source	Battery (needs frequent charging)	Battery (needs frequent charging)	Solar panels + battery	Solar panels + battery, optimized for Saudi climate
User Monitoring	Manual/None	Real-time health monitoring via IoT	Manual/None	Real-time health and location monitoring via IoT
Navigation	Manual joystick	Joystick, voice, gesture, or app control	Manual joystick	Multi-mode: joystick, app, gesture, voice, GPS
Obstacle Detection	Basic (if any)	Advanced (ultrasonic, IR, sensors)	Basic (if any)	Advanced (ultrasonic, IR, AI-based)
Energy Efficiency	Low	Low	High (solar)	Very high (solar + smart management)
Sustainability	Low	Low	Medium	High (renewable energy, eco-friendly)
Suitability for Saudi Arabia	Limited by charging infrastructure	Limited by charging infrastructure	Good (solar abundant)	Excellent (solar optimized, IoT for local needs)

2.5 Sensors and Communication

In designing a smart wheelchair system, the choice of sensors and communication technologies is critical for ensuring user safety, accurate navigation, reliable connectivity, and energy efficiency. The table below presents a concise comparison of key technologies commonly used in assistive mobility systems, including their core functions, strengths, and potential limitations. This evaluation supports informed component selection that aligns with the environmental and infrastructural needs of Saudi Arabia, where high temperatures, dust, and connectivity gaps may affect system performance.

Table 2.2: Comparative Table: Sensors and Communication Technologies in Smart Wheelchairs

Technology	Functionality	Advantages	Limitations
Ultrasonic Sensors	Obstacle detection	Accurate, low-cost, effective in various lighting	Limited range, affected by soft materials
Infrared Sensors	Proximity and obstacle detection	Fast response, low power, simple integration	Affected by direct sunlight, limited range
GPS Module	Location tracking	Real-time outdoor navigation, geofencing	Poor performance indoors, needs clear sky access
Bluetooth	Wireless communication	Low energy, good for short-range device pairing	Short range, prone to interference
Wi-Fi	Data transmission	High-speed communication, remote monitoring	High power usage, depends on infrastructure
Solar Panel	Power supply	Sustainable, cost-effective, eco-friendly	Performance depends on sunlight availability

2.5.1 Key Literature Review

2.5.1.1 Comparative Review of Key Literature Sources

A wide range of studies has contributed to the development of IoT-enabled and solar-powered smart wheelchair technologies. These works vary in their technical scope, research focus, and real-world applicability.

To better understand the current research landscape and identify innovation gaps, the following table summarizes key publications relevant to this project. Each source is evaluated based on its title, authorship, and core findings or contributions. This comparative review helps position the proposed solution in the broader context of assistive mobility research and supports the justification for its novelty and relevance.

Table 2.3: Comparative table of key sources on IoT-enabled and solar-powered smart wheelchairs

Reference	Title/Author	Description/Findings
[5]	Bhatnagar et al.	Reviews design of solar-powered wheelchairs, highlighting energy efficiency and user autonomy.
[6]	Roy	Proposes an IoT-based health monitoring system with remote support for disabled users.
[7]	Sreevatsav et al.	Presents IoT smart wheelchair with GPS and obstacle avoidance for mobility enhancement.
[8]	Amer et al.	Develops eye-gaze control system, improving accessibility for severely impaired users.
[9]	Riaz and Aamir	Designs solar-powered wheelchair using PV cells for independent operation.

2.5.1.2 Comparison of Related Smart Wheelchair Solutions

To further support the design and justification of the proposed smart wheelchair system, it is important to evaluate existing solutions in the field. Several prior works have attempted to integrate technologies such as IoT, solar power, and adaptive control mechanisms into assistive mobility devices. However, each of these projects varies in terms of feature completeness, technical limitations, and contextual relevance. The table below provides a structured comparison of selected smart wheelchair projects from recent literature. Each solution is assessed based on its core features, notable limitations, and its relevance or influence on the current project. This comparison helps to identify gaps in existing technologies and demonstrates how the proposed system aims to bridge those gaps.

Table 2.4: Comparison of Related Smart Wheelchair Solutions

Reference	Features	Limitations	Relevance to Project
[7]	IoT, GPS, obstacle sensors	Grid-dependent, no health tracking	Inspired navigation and safety
[8]	Eye-gaze control interface	No solar or IoT integration	Enhanced accessibility strategy
[9]	Solar-powered drive with PV panels	No real-time monitoring or IoT	Informed energy architecture
[10]	Solar + health + obstacle detection	Limited user interface options	Validated hybrid sensing system

2.5.1.3 Comparative Table of Additional Relevant Studies

Beyond the core literature, several additional studies have explored advanced features in smart wheelchair technologies. These works focus on integrating IoT, renewable energy, and intelligent control mechanisms, aiming to improve mobility, independence, and usability for disabled users. The following table summarizes key findings from recent publications that demonstrate practical innovations in design, energy efficiency, user interfaces, and health monitoring. Reviewing these contributions supports a deeper understanding of the current state of the field and reinforces the relevance of the proposed project.

Table 2.5: Comparative table of additional relevant studies

Reference	Title/Author	Description/Findings
[10]	Aswathy et al.	Introduces solar-powered wheelchair equipped with health and obstacle sensors, focusing on user safety, IoT-based monitoring, and energy sustainability.
[11]	Metkari et al.	Designs a cost-effective solar-powered wheelchair adapted for local environmental conditions, emphasizing fabrication methods and real-world testing.
[12]	Sadi et al.	Develops a gesture-controlled smart wheelchair using IoT and computer vision to enable intuitive, hands-free mobility for physically disabled users.
[13]	Cui et al.	Proposes a multi-mode IoT-based wheelchair integrating cloud connectivity, gesture recognition, and real-time health data collection through advanced sensors.

Chapter 3

METHODOLOGY

3.1 Introduction

This chapter outlines the development and implementation methodology for the IoT-enabled, solar-powered smart wheelchair. The primary goal of this project is to enhance mobility, safety, and energy efficiency for individuals with physical disabilities, particularly in regions with limited infrastructure and harsh environmental conditions such as Saudi Arabia. Unlike traditional mobility aids, the proposed solution integrates Internet of Things (IoT) technologies with renewable solar energy to enable real-time monitoring, smart navigation, and off-grid operational sustainability.

The adopted methodology is user-centered and engineering-oriented. It emphasizes modularity and scalability to support future extensions such as additional sensors, AI-assisted analytics, and enhanced energy optimization. The chapter is structured to achieve the following objectives:

- **Smart Connectivity and Real-Time Monitoring:** Deploying IoT sensors for continuous monitoring, obstacle awareness, and safer navigation.
- **Energy Efficiency through Solar Integration:** Leveraging local solar resources to reduce dependence on the electrical grid and increase reliability.
- **Advanced Control Mechanisms:** Supporting intuitive control with responsive motion and safety triggers.
- **Data Communication and Cloud Connectivity:** Using lightweight messaging (e.g., MQTT) to

enable real-time data exchange, remote monitoring, and alerts.

3.2 Methodological Overview

The project follows an iterative methodology that starts from requirement identification and proceeds through modeling, component selection, prototyping, and evaluation. Iteration is essential because assistive mobility systems are strongly affected by real-world conditions (terrain, user behavior, and sensor noise). Each development cycle improves safety and usability based on test results and feedback.

3.3 Research and Project Design

The primary objective of this research is to design, develop, and evaluate a smart wheelchair system that leverages IoT connectivity and solar power to enhance user safety, autonomy, and energy efficiency. The design focuses on:

- **Remote monitoring:** enabling caregivers/medical staff to view status and key indicators.
- **Obstacle detection:** reducing collision risk in indoor/outdoor environments.
- **Energy sustainability:** increasing daily operating time using solar harvesting and battery storage.

3.3.1 Why these diagrams were selected

The following diagrams were selected because they explain the system behavior clearly without overwhelming the reader with low-level details:

- **Component distribution (Fig. 3.1):** communicates system priorities and design focus.
- **Data Flow Diagram (Fig. 3.2):** explains how data moves from sensors to cloud/users and how control commands return.
- **Power management flow (Fig. 3.3):** clarifies energy generation, regulation, storage, and delivery to the load.

This makes the chapter readable for both technical and non-technical reviewers.

Distribution of Key Components - IoT-Enabled Solar-Powered Smart Wheelchair

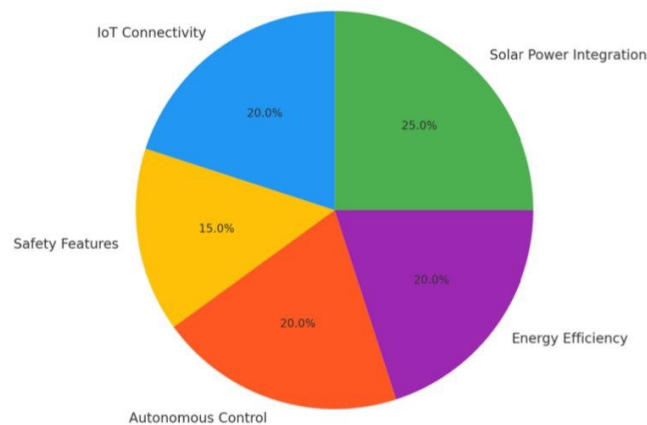


Figure 3.1: Distribution of Key Components in the IoT-Enabled Solar-Powered Smart Wheelchair.

Figure 3.1 provides a high-level overview of the smart wheelchair architecture. The diagram highlights that the system design is driven primarily by sustainability and safety. Solar energy and battery storage form the energy backbone of the system, while IoT connectivity enables continuous data exchange for monitoring and control. Sensing and control modules ensure safe navigation, and the user interaction layer focuses on accessibility and ease of use for individuals with physical disabilities.

3.3.2 Data Flow Diagram (DFD)

The Data Flow Diagram (DFD) represents the real-time flow of data within the system. It illustrates how data is collected from sensors, processed by the microcontroller, transmitted to the cloud, and displayed on the mobile or web interface for monitoring and control.

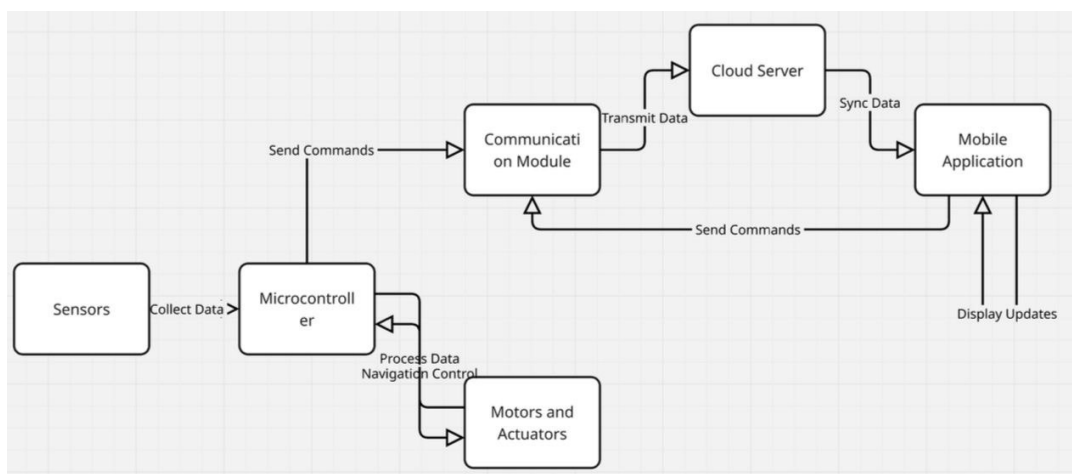


Figure 3.2: Data Flow Diagram (DFD) for IoT-Enabled Solar-Powered Smart Wheelchair.

Figure 3.2 explains how data flows across the system components. Sensor data such as obstacle distance, health indicators, and location information are processed locally by the microcontroller. The processed data is then transmitted to the cloud platform, where it can be visualized through a user interface or accessed by caregivers. Control commands follow the reverse path, ensuring real-time responsiveness.

3.3.3 Power Management Flow Diagram

The power management flow diagram illustrates the end-to-end flow of energy generation, storage, and distribution across the system. It ensures stable and reliable operation under variable sunlight conditions.

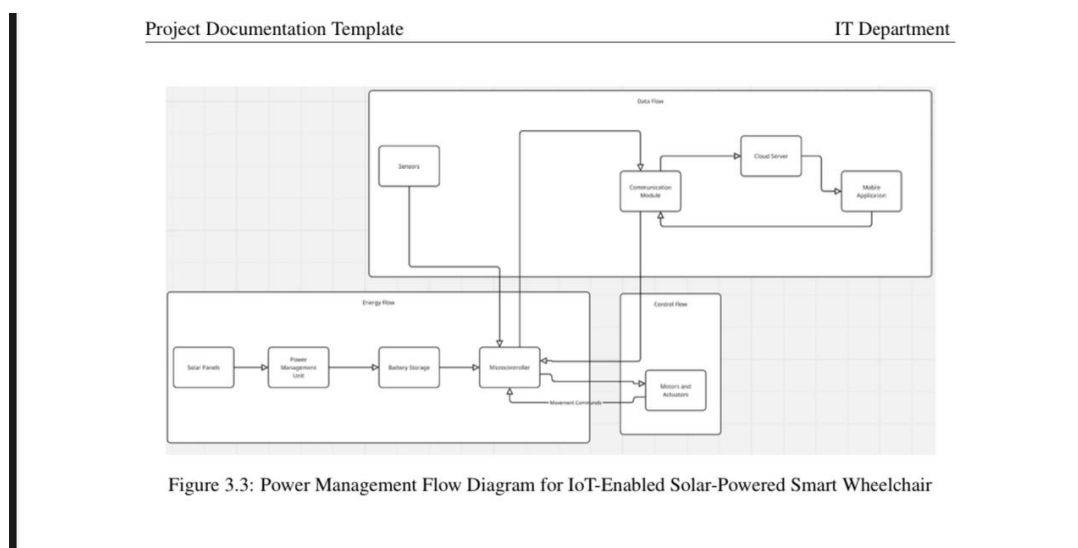


Figure 3.3: Power Management Flow Diagram for IoT-Enabled Solar-Powered Smart Wheelchair

Figure 3.3: Power Management Flow Diagram for IoT-Enabled Solar-Powered Smart Wheelchair.

As shown in Figure 3.3, solar energy is harvested and regulated before being stored in the battery. The stored energy is then distributed to critical components such as the control unit, sensors, and motors, enabling off-grid operation and reducing dependence on external charging sources.

3.4 System Development Process and Data Collection

Development was conducted through iterative integration:

- **Requirements analysis:** identifying system needs from literature and stakeholder expectations.
- **Hardware/software integration:** connecting sensors, solar module, control unit, and communication layer.

- **Prototype testing:** verifying functionality under controlled conditions, then refining for reliability.

3.4.1 Ultrasonic Sensor Data Collection

Obstacle detection relies on ultrasonic measurement. The sensor sends a trigger pulse and measures the echo return time to compute distance. This distance is used in real-time to prevent collisions and to support safe navigation decisions.

```
// Define pins for Ultrasonic Sensor
const int trigPin = 9;
const int echoPin = 10;

void setup() {
    Serial.begin(9600);           // Start the serial communication
    pinMode(trigPin, OUTPUT);     // Set the trigger pin as output
    pinMode(echoPin, INPUT);      // Set the echo pin as input
}

void loop() {
    // Send a short pulse to trigger the sensor
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);

    // Receive the echo and calculate the distance
    long duration = pulseIn(echoPin, HIGH);
    float distance = duration * 0.034 / 2;

    // Display the distance in the serial monitor
    Serial.print("Distance: ");
    Serial.print(distance);
    Serial.println(" cm");

    // Small delay for stability
    delay(500);
}
```

Figure 3.4: Sample Code for Ultrasonic Sensor Distance Measurement (Illustration).

3.4.2 Ultrasonic Sensor Code (written in LaTeX)

To keep the report self-contained, Listing 3.1 provides a clear implementation example of distance measurement logic.

Listing 3.1: Ultrasonic Sensor Distance Measurement

```
const int trigPin = 9;
const int echoPin = 10;

void setup() {
    Serial.begin(9600);
    pinMode(trigPin , OUTPUT);
    pinMode(echoPin , INPUT);
}

void loop() {
    // Trigger pulse
    digitalWrite(trigPin , LOW);
    delayMicroseconds(2);
    digitalWrite(trigPin , HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin , LOW);

    // Echo duration -> distance
    long duration = pulseIn(echoPin , HIGH);
    float distance = duration * 0.034 / 2;

    Serial.print("Distance:■");
    Serial.print(distance);
    Serial.println("■cm");

    delay(500);
}
```

3.4.3 Pseudocode for Ultrasonic Sensor Data Collection

The pseudocode below summarizes the measurement process at a higher level, which helps readers understand the logic even without embedded programming experience.

Pseudocode for Ultrasonic Sensor Data Collection

- **Initialize Pins:**
 - trigPin = 9 (*Set the Trigger pin*)
 - echoPin = 10 (*Set the Echo pin*)
- **Setup Configuration:**
 - Start Serial Communication at 9600 baud.
 - Set trigPin as OUTPUT.
 - Set echoPin as INPUT.
- **Main Loop:**
 - Send a LOW pulse to trigPin.
 - Wait for 2 microseconds.
 - Send a HIGH pulse to trigPin.
 - Wait for 10 microseconds.
 - Send a LOW pulse to trigPin.
- **Measure Echo Duration:**
 - Read the duration of the pulse from echoPin.
- **Calculate Distance:**
$$\text{Distance} = \frac{\text{Duration} \times 0.034}{2}$$
- **Display Distance:**
 - Print the distance to the serial monitor.
- **Stability Delay:**
 - Wait for 500 milliseconds.

Figure 3.5: Pseudocode for Ultrasonic Sensor Data Collection.

The measurement sequence can be summarized as follows: initialize pins, generate a trigger pulse,

measure the echo duration, and compute distance using the standard ultrasonic formula:

$$\text{Distance (cm)} = \frac{\text{Duration} \times 0.034}{2}$$

3.5 MQTT Communication for Real-time Data

MQTT was selected because it is lightweight, efficient, and widely adopted in IoT applications. Its publish/subscribe model supports real-time telemetry and remote monitoring using minimal bandwidth and computational resources.

3.5.1 MQTT Code Example (written in LaTeX)

Listing 3.2 demonstrates a simplified MQTT workflow: connect to Wi-Fi, configure the broker, and publish periodic status messages.

Listing 3.2: MQTT Communication Example for Real-time Monitoring

```
#include <WiFi.h>
#include <PubSubClient.h>

const char* ssid      = "Your_SSID";
const char* password  = "Your_PASSWORD";
const char* mqttServer = "192.168.1.10";
const int   mqttPort  = 1883;

WiFiClient espClient;
PubSubClient client(espClient);

void setup() {
    Serial.begin(115200);

    WiFi.begin(ssid, password);

    while (WiFi.status() != WL_CONNECTED) {
```

```
    delay (500);
    Serial.println("Connecting to WiFi...");
}
Serial.println("WiFi connected");

client.setServer(mqttServer, mqttPort);
Serial.println("MQTT server configured");
}

void loop() {
    if (!client.connected()) {
        client.connect("WheelchairClient");
    }
    client.loop();

    // Publish status
    client.publish("wheelchair/status", "Operational");
    delay(2000);
}
```

This communication layer enables sending sensor readings and receiving commands (e.g., emergency stop). It also supports logging and analytics at the cloud side, which can be used for maintenance and performance improvements.

3.6 Materials and Tools Selection

This section summarizes the materials used in the prototype. For each selection, a simple justification and a deeper technical justification are provided.

3.6.1 Hardware Materials

Component	Simple Explanation	Technical Justification
Microcontroller (ESP32 / Arduino)	Controls the wheelchair and connects sensors.	Offers reliable embedded control, multiple GPIOs, and supports IoT networking stacks (Wi-Fi/MQTT) depending on the selected board.
Ultrasonic Sensor	Detects obstacles in front of the wheelchair.	Enables real-time distance measurement with low latency, supporting collision avoidance and safety thresholds.
Solar Panel (5V)	Generates power from sunlight.	Matches the local environment (high sun exposure) and reduces dependence on external charging.
Power Regulation (PMU/Converter)	Stabilizes voltage for safe operation.	Protects battery and electronics by regulating solar input and improving energy efficiency and stability.
Battery Storage	Stores energy for continuous operation.	Buffers intermittent solar input and supports peak motor current demands.
Motor Driver + DC Motors	Moves the wheelchair.	Provides controllable speed/direction and supports safe, responsive mobility.

Table 3.1: Hardware Components Used in the Smart Wheelchair Prototype.

3.6.2 Software Tools

Software Tool	Simple Explanation	Technical Justification
Arduino IDE / PlatformIO	Used to program the microcontroller.	Supports rapid prototyping, serial debugging, and integration with IoT and sensor libraries.
MQTT Protocol	Transfers data between wheelchair and cloud.	Lightweight pub/sub model minimizes bandwidth usage and supports scalable monitoring.
Web-Based Dashboard (UI)	Shows data clearly for users.	Improves accessibility and UX through live updates, alerts, and simplified control interface.

Table 3.2: Software Tools and Technologies Used in the System.

3.7 System Testing and Evaluation Plan

Testing was designed to verify functional correctness, performance, and usability. The plan includes controlled tests and observation-based feedback sessions.

Test Type	Description	Expected Outcome
Obstacle Detection Test	Evaluate distance accuracy and safety response.	Accurate detection within expected range and safe response with minimal delay.
Communication Test (MQTT)	Publish updates and receive commands.	Stable connection, timely updates, and correct command execution.
Power Stability Test	Observe solar charging and battery discharge behavior.	Stable voltage to controller and measurable charge under sunlight.
Usability Test (UX)	Assess clarity and ease of use for end users/caregivers.	Users can understand status quickly and operate controls consistently.

Table 3.3: System Testing and Evaluation Summary.

3.8 Ethical Considerations

Ethical principles were followed to protect participants and ensure accessibility:

- **Informed consent** from participants involved in demonstrations or feedback sessions.
- **Privacy and confidentiality** for any health-related or identity-related data.
- **Accessibility** in the user interface and interaction design to avoid discrimination.

3.9 Conclusion

This chapter presented the methodology used to design a sustainable, safe, and user-centered IoT-enabled solar-powered smart wheelchair. The diagrams clarify system structure, data flow, and power flow, while the development process describes how sensing, control, communication, and energy management were integrated. The provided materials and testing plan support reliability and practical evaluation. The next chapter presents the implementation details of the prototype.

Chapter 4

IMPLEMENTATION AND TESTING

4.1 Introduction

This chapter presents the practical implementation and testing of the IoT-Enabled Solar-Powered Smart Wheelchair. It focuses on translating the theoretical design into a functional prototype by integrating hardware components, embedded software, and the user interface.

The implementation emphasizes real-world feasibility, simplicity, and safety. All components were selected and assembled to demonstrate a working prototype that reflects the core objectives of the project: smart mobility, user safety, and sustainable energy usage.

4.2 System Overview

The implemented system consists of three main layers:

- **Hardware Layer:** Sensors, motors, power unit, and controller.
- **Software Layer:** Embedded firmware and control logic.
- **Interface Layer:** Web-based dashboard for monitoring and control.

This layered structure improves reliability and simplifies maintenance and future upgrades.

4.3 Physical Prototype Implementation

4.3.1 Overall Wheelchair Assembly (Top View)

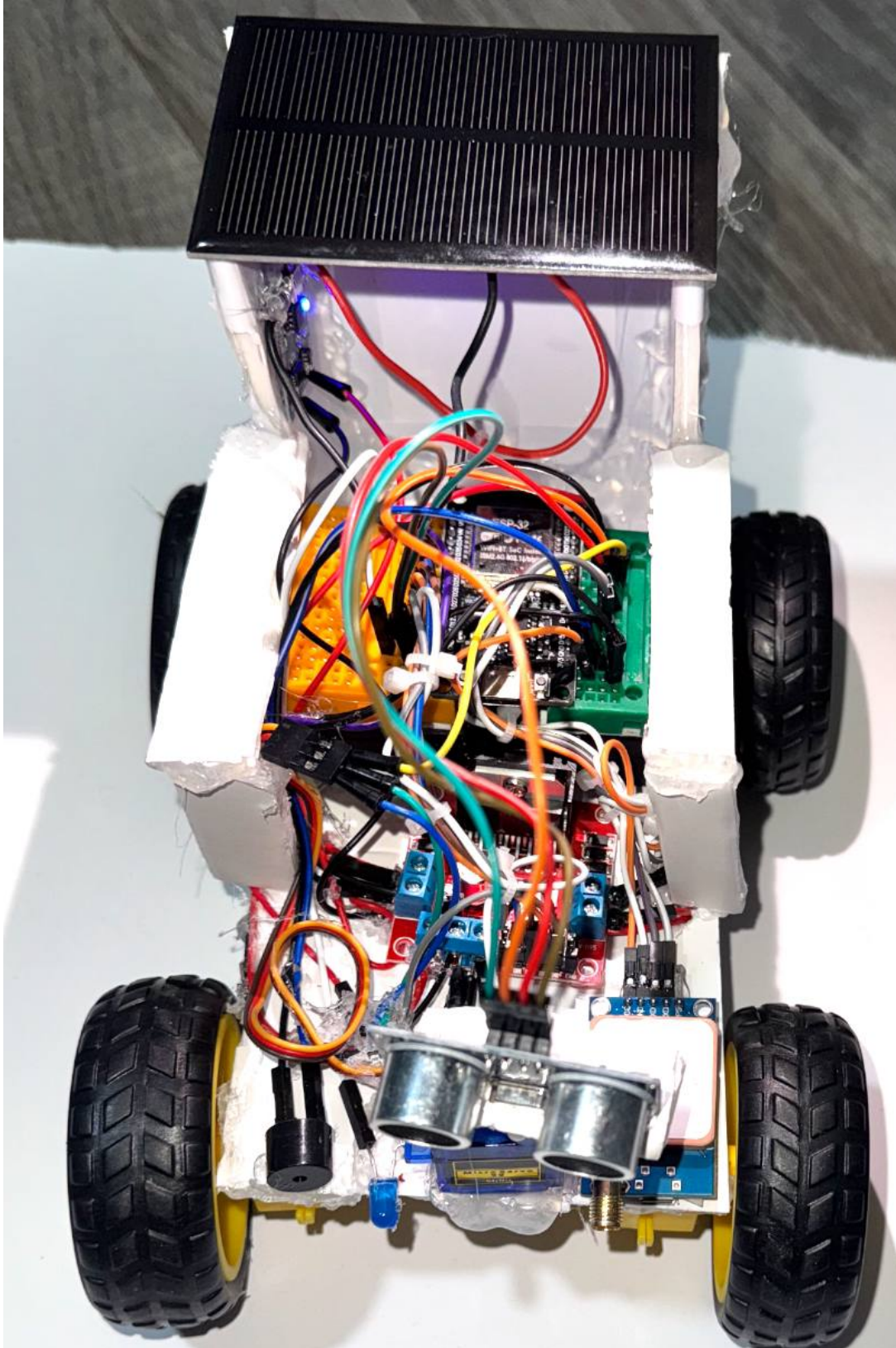


Figure 4.1: Top View of the Smart Wheelchair Showing Integrated Hardware Components.

Figure 4.1 illustrates the complete assembly of the smart wheelchair, including the ESP32 controller, motor driver, ultrasonic sensor, GPS module, pulse sensor, battery pack, and solar panel placement.

4.3.2 Hardware Close-Up View

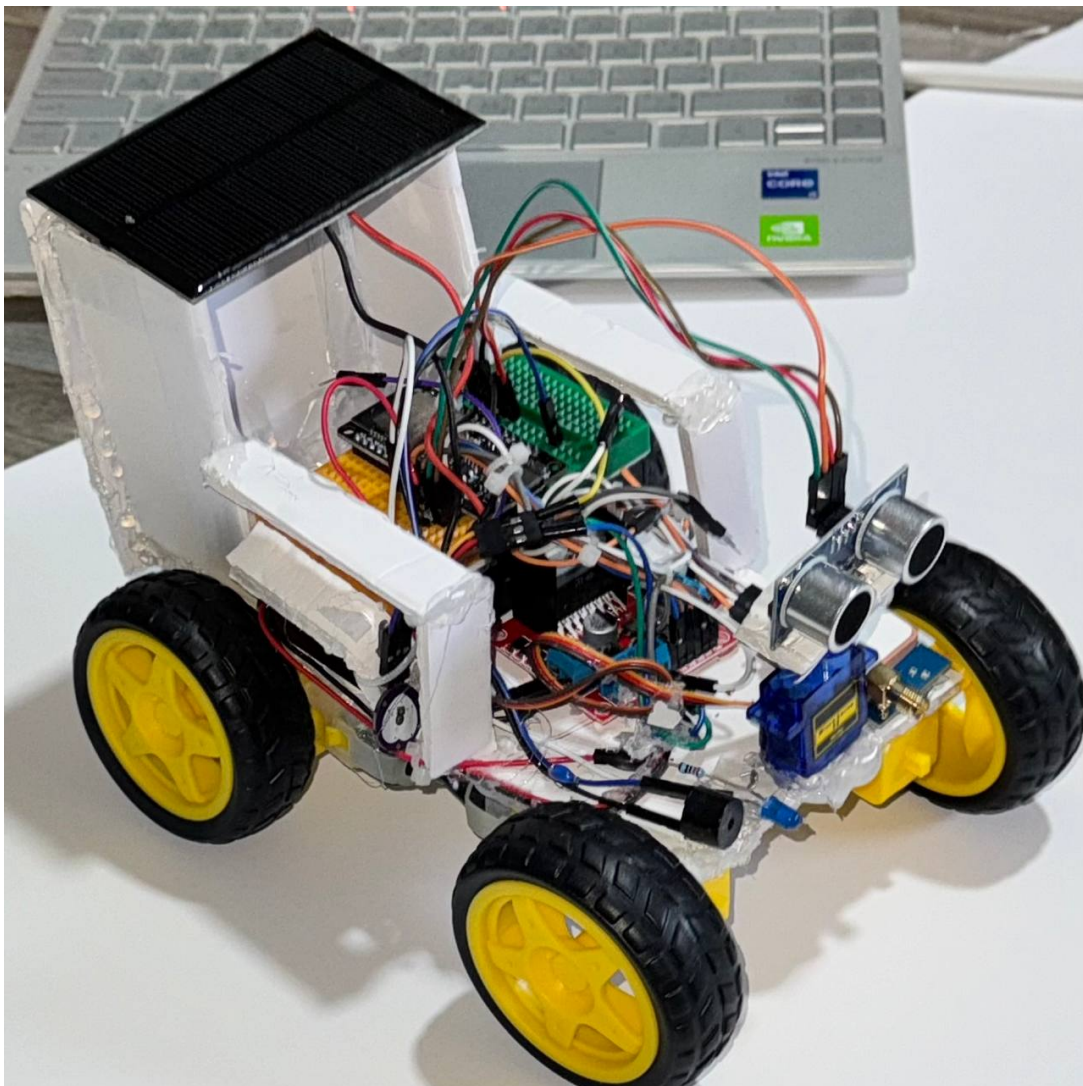


Figure 4.2: Close-Up View of the Hardware Components and Wiring.

This figure highlights the internal wiring, ESP32 board, L298N motor driver, and sensor connections, demonstrating the compact and modular hardware design.

4.3.3 Solar Power Integration

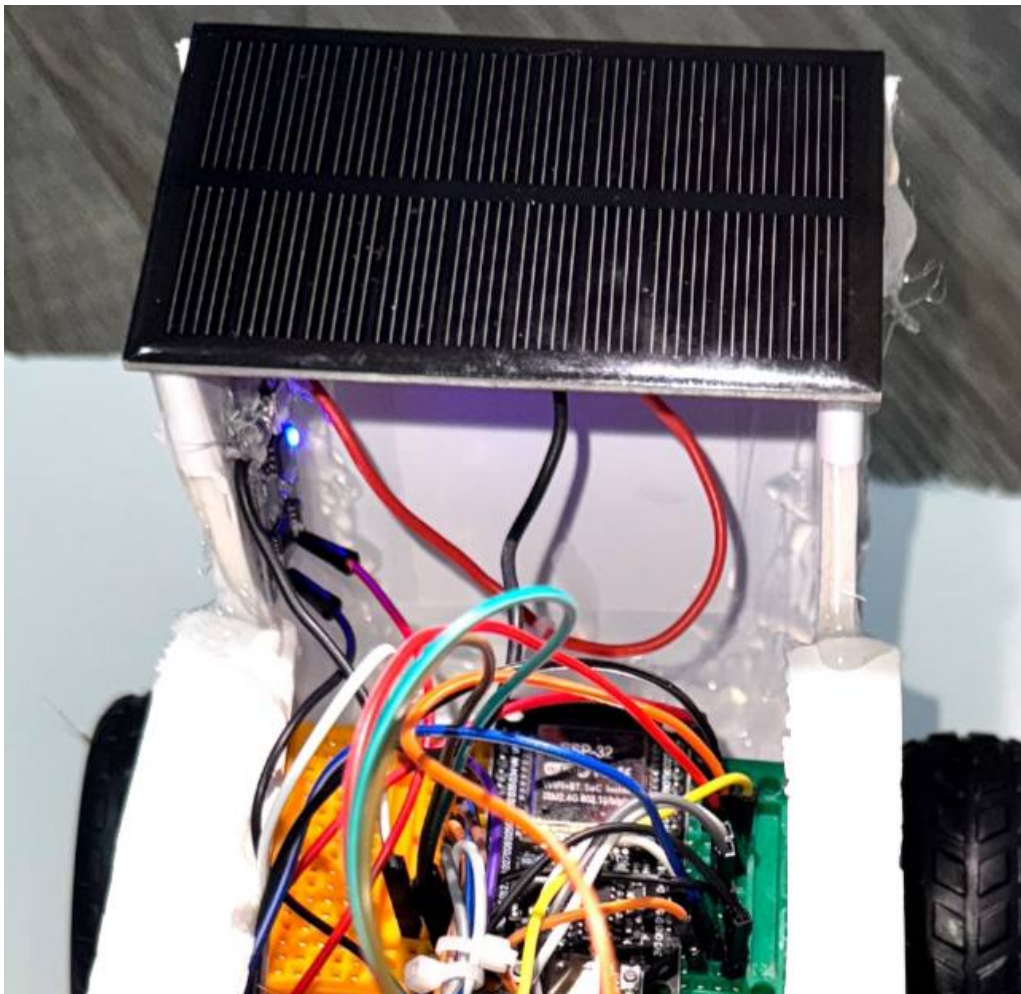
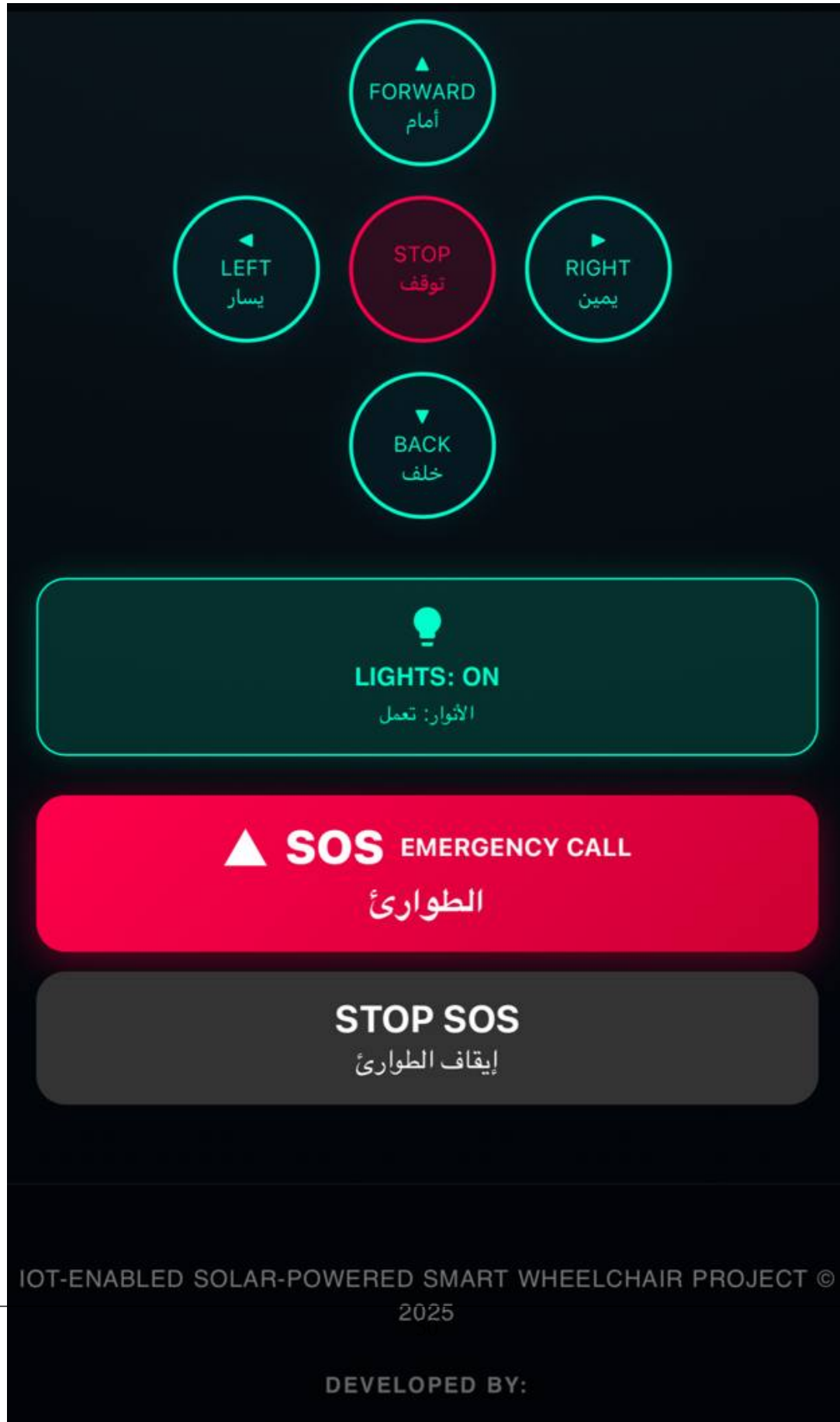


Figure 4.3: Solar Panel and Power Management Integration.

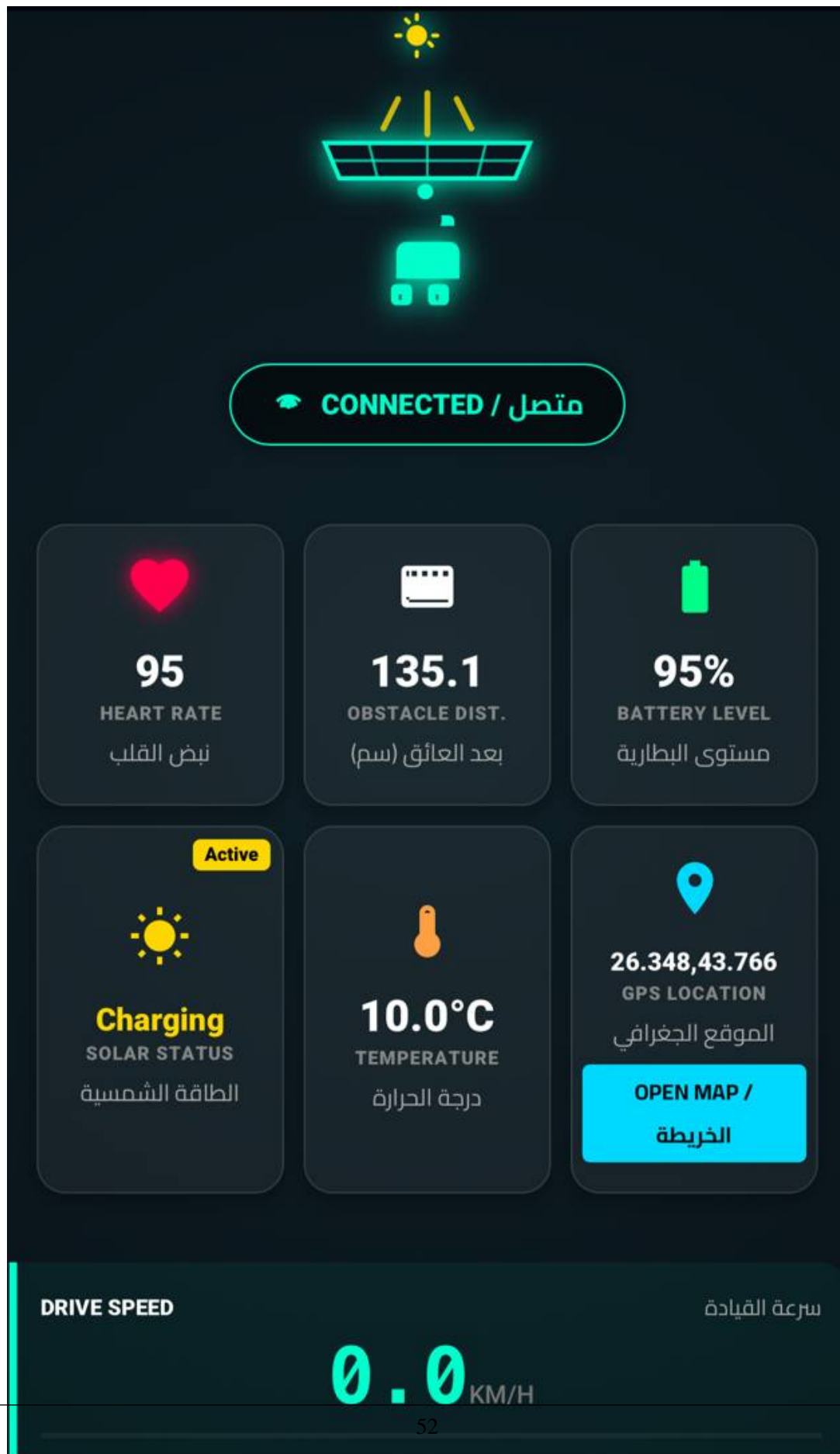
The solar panel supplies renewable energy to the system through a charging and regulation module, ensuring extended operation and energy sustainability.

4.4 User Interface Implementation

4.4.1 Dashboard Control Interface



4.4.2 Dashboard Monitoring Interface



The dashboard enables real-time monitoring of obstacle distance, heart rate, battery level, temperature, and GPS location, improving user safety and remote supervision.

4.5 System Diagrams

4.5.1 Use Case Diagram

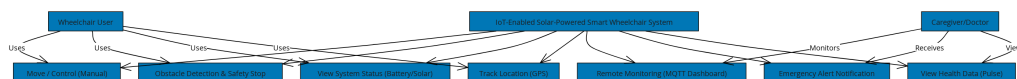


Figure 4.6: Use Case Diagram for the IoT-Enabled Smart Wheelchair.

4.5.2 Class Diagram

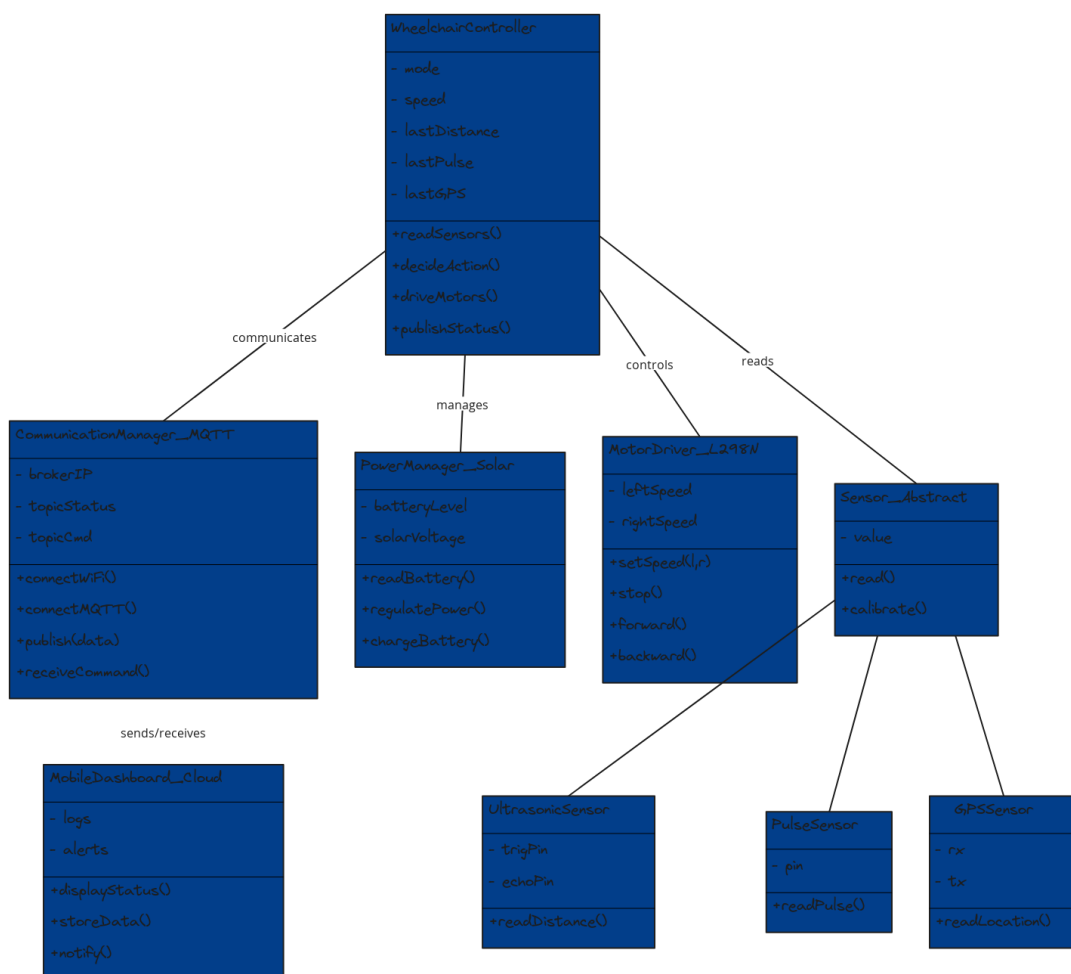


Figure 4.7: Class Diagram of the Smart Wheelchair Software Architecture.

4.6 Testing and Validation

4.6.1 Functional Testing

The prototype was tested to verify correct system behavior, including movement control, obstacle detection, and wireless communication.

Test Item	Expected Result	Status
Obstacle Detection	Accurate distance measurement	Passed
Motor Control	Smooth directional movement	Passed
Web Dashboard	Real-time data update	Passed
Emergency Alert	Audible and visual alert	Passed

Table 4.1: Implementation Testing Results Summary.

4.7 Conclusion

This chapter demonstrated the successful implementation and testing of the IoT-enabled solar-powered smart wheelchair. The prototype validates the practical feasibility of integrating IoT technologies, renewable energy, and assistive mobility into a single functional system.

Chapter 5

RESULTS AND DISCUSSIONS

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5.1 Introduction

This chapter presents the experimental results obtained from implementing and testing the IoT-Enabled Solar-Powered Smart Wheelchair prototype. The focus is to validate the practical behavior of the system in real scenarios, including obstacle detection performance, dashboard monitoring reliability, lighting functionality, connection-loss behavior, and emergency (SOS) safety features.

The results are presented using real prototype screenshots and test observations. Each result is followed by a short discussion explaining what the result means, why it matters for user safety, and what limitations were observed.

5.2 Obstacle Detection and Safety Response Results

Obstacle detection is a core safety function in the proposed wheelchair system. The ultrasonic sensor continuously measures the distance to objects in front of the chair. When the measured distance falls below a defined threshold, the system triggers a safety response (warning/stop/avoidance depending on configuration).

Figures 5.1a–5.1i show sample results during real testing when an obstacle was placed near the

wheelchair and moved across its path. The dashboard also reflects the distance value in real time, allowing the user/caregiver to verify the safety status.



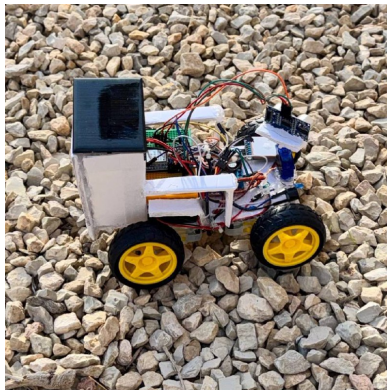
(a) Obstacle detected (initial).



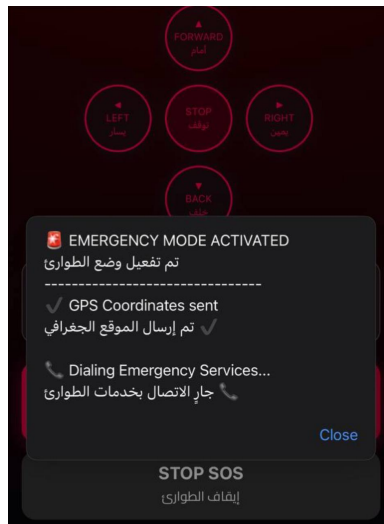
(b) Distance value updates.



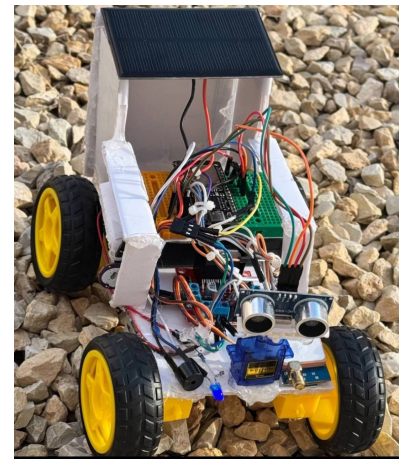
(c) Object approaching chair.



(d) Threshold region reached.



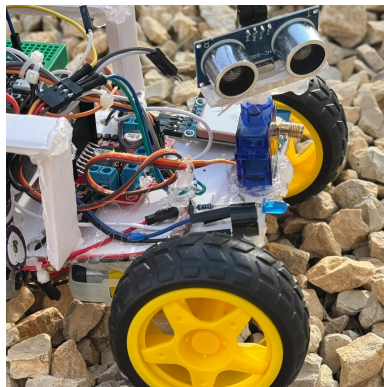
(e) Safety feedback shown.



(f) Obstacle moved sideways.



(g) Distance increases again.



(h) Clear path confirmed.



(i) Stable readings after removal.

Figure 5.1: Obstacle detection test sequence using the ultrasonic sensor and real-time dashboard feed-back.

Discussion: The results confirm that the sensor reading updates fast enough to support practical collision avoidance. The system can detect an approaching obstacle, show the distance clearly on the dashboard, and trigger safety behavior when the distance becomes unsafe. In real environments, accuracy can be affected by sensor angle, reflective surfaces, and noise, so physical mounting and filtering are important for stable readings.

5.3 Solar Power and Battery Status Observations

Although the prototype is simplified, the solar-powered design goal is validated by monitoring battery level and solar charging status indicators on the interface. The system uses a solar charging module to charge lithium-ion batteries and supply stable power to the ESP32 and connected sensors.

Discussion: In high-sun regions (e.g., Saudi Arabia), solar charging supports longer daily operation. However, the charging rate depends on sunlight intensity, panel angle, and total load. In a full-scale wheelchair, power sizing and safety protection (e.g., battery management, overcurrent protection) must be engineered more strongly.

5.4 Lighting System Result (Visibility Enhancement)

The prototype includes an external lighting feature to improve visibility and safety, especially in low-light or night conditions. Figure 5.2 shows the lights activated during testing.

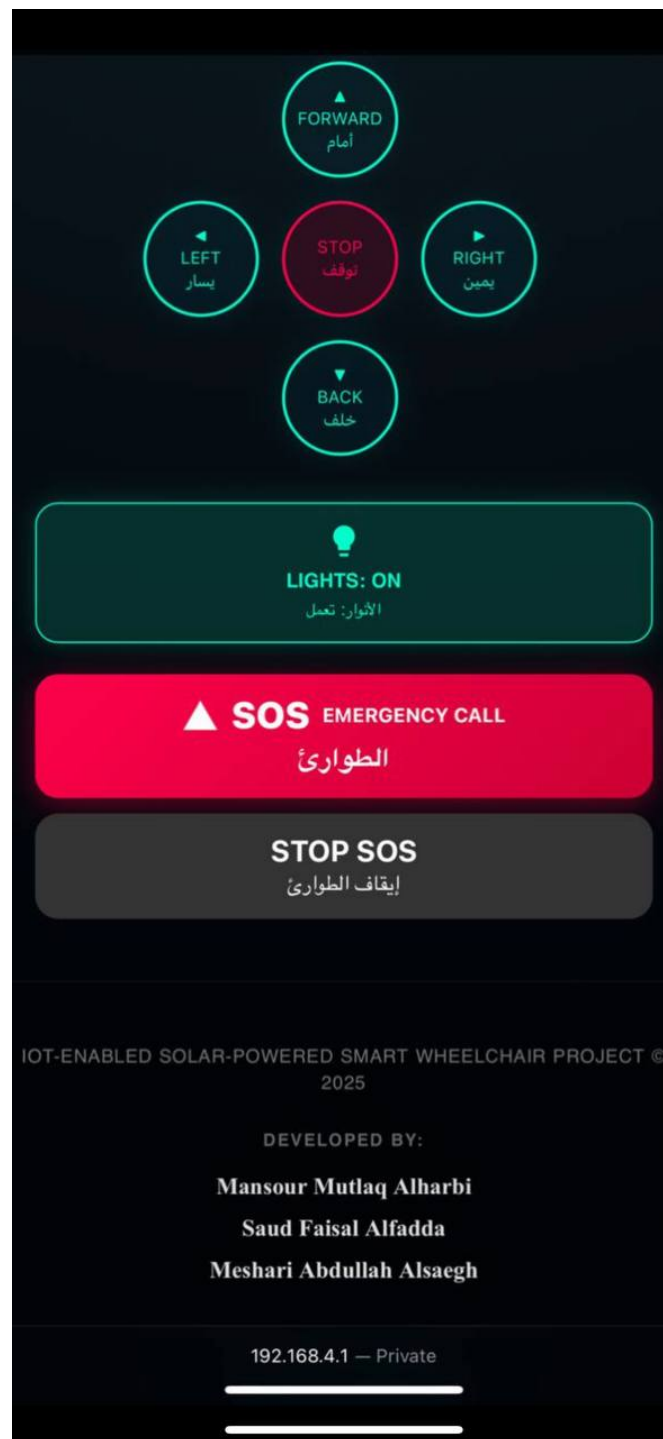


Figure 5.2: Lighting system activated to improve visibility and safety.

Discussion: The lighting feature is simple but useful in practice. It helps others notice the wheelchair in dark areas and supports safer navigation. For a full product, the lights can be integrated with automatic conditions (e.g., light sensor) or emergency flashing mode.

5.5 Disconnected Mode and Reliability Behavior

The wheelchair uses a Wi-Fi access point hosted on the ESP32 for local control. When the user moves away, loses Wi-Fi, or the device disconnects, the interface should reflect connection loss and the system should behave safely (e.g., stop movement after a timeout).

Figures 5.3a–5.3c show sample results during disconnection testing.

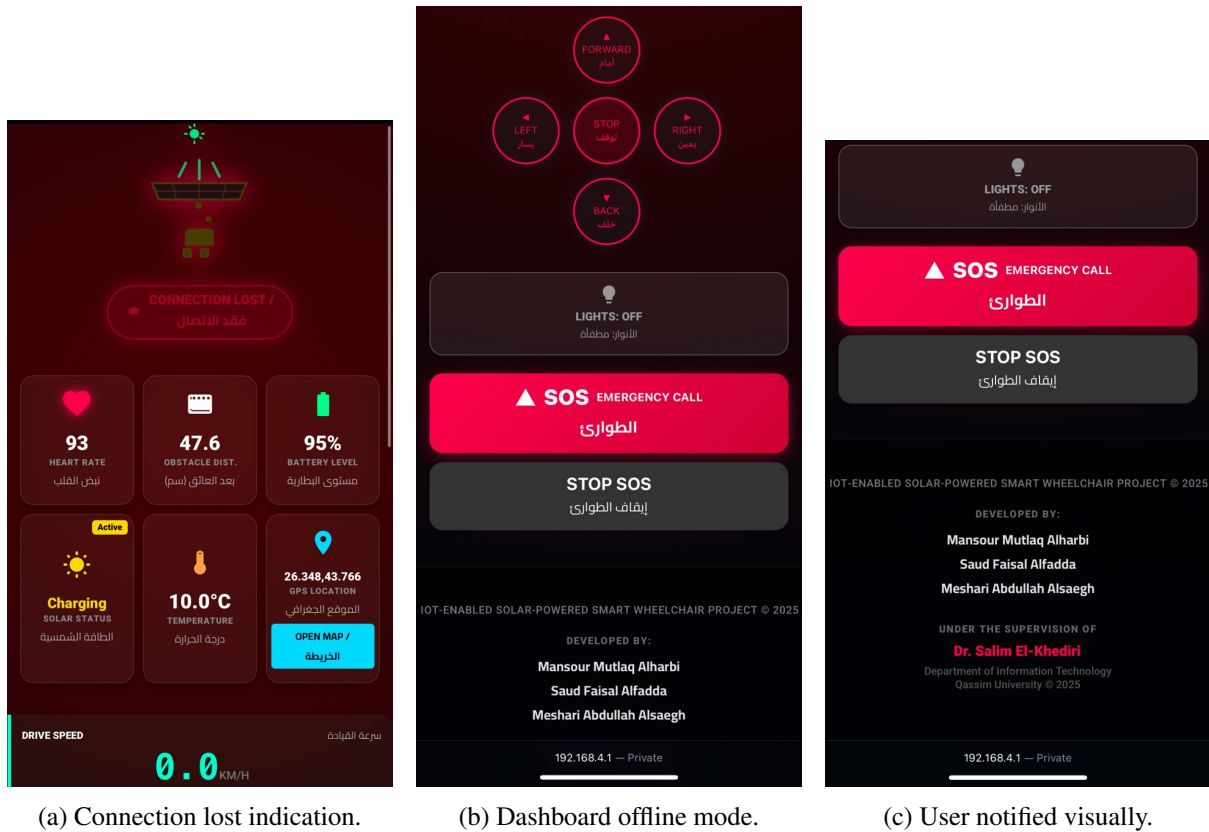
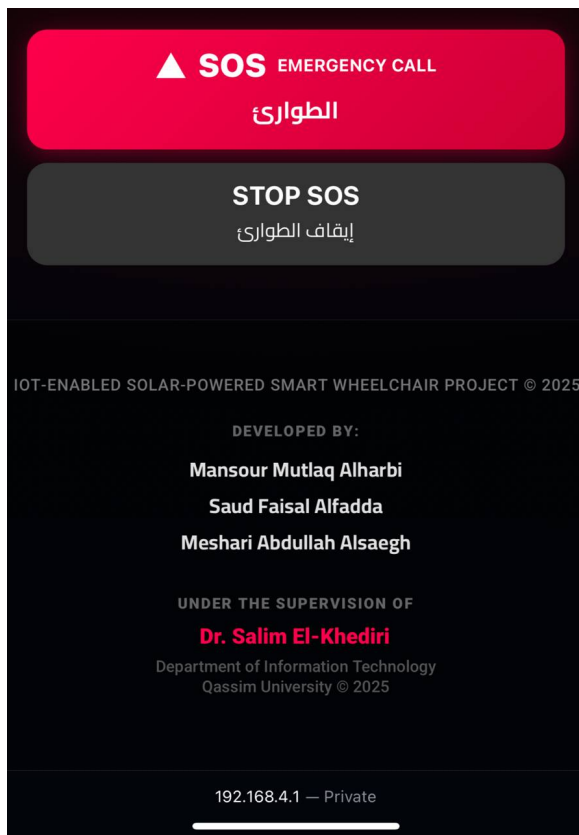


Figure 5.3: Disconnected mode results (connection loss and user feedback).

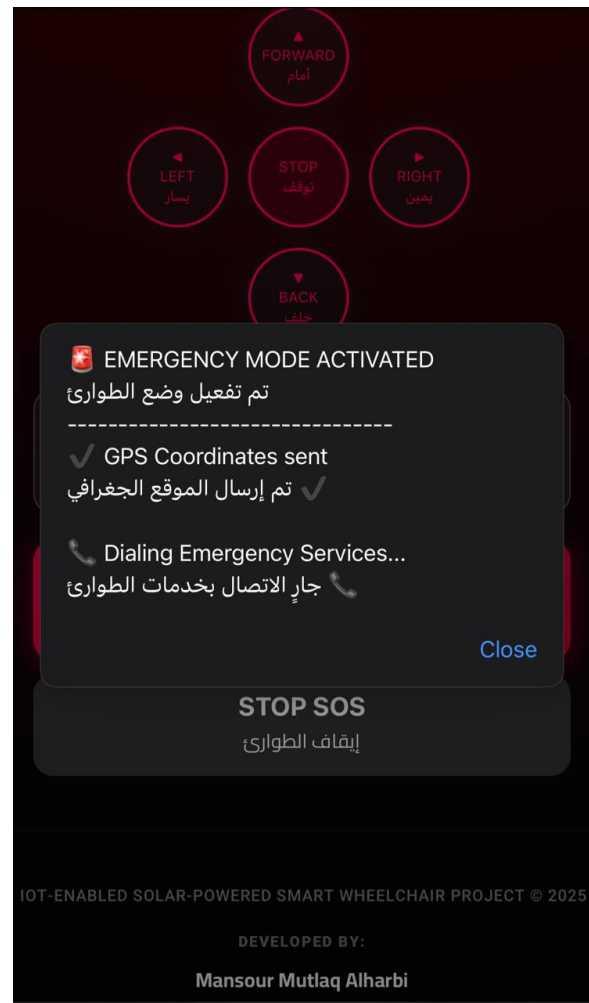
Discussion: These results confirm that the user receives clear feedback when Wi-Fi is lost. More importantly, the firmware safety rule that stops the motors after a connection timeout reduces risk. In future versions, BLE fallback or cloud relay can be added for higher reliability.

5.6 Emergency SOS Feature Results

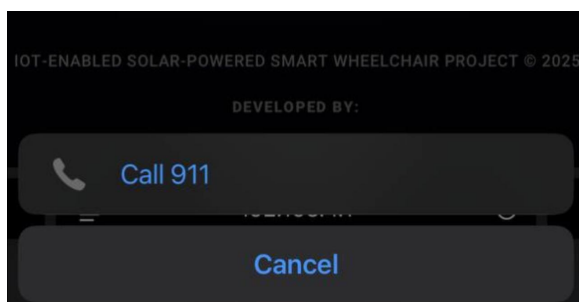
The SOS feature is designed as a critical safety function. When the user activates SOS, the system triggers a buzzer and warning indicators, and the dashboard provides clear confirmation. Figures 5.4a–5.4d show the SOS activation and stopping sequence.



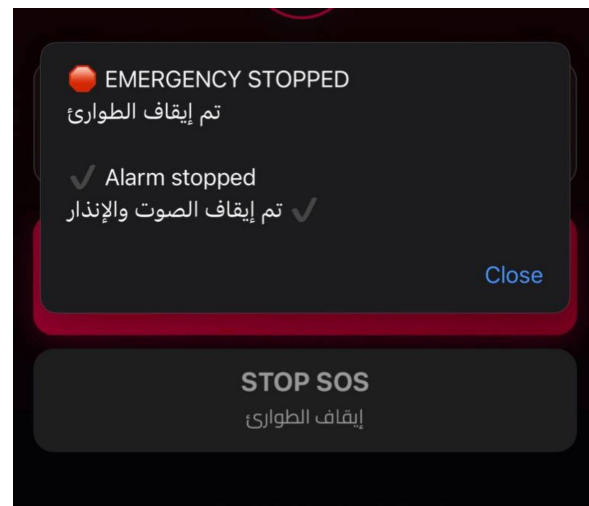
(a) SOS button available on dashboard.



(b) SOS activated (alert mode).



(c) Emergency feedback and warning.



(d) STOP SOS used to cancel emergency.

Figure 5.4: Emergency SOS feature results (activation and stop sequence).

Discussion: The SOS workflow improves safety because it allows the user to request help quickly and provides strong feedback through sound/light. The STOP SOS action is equally important to prevent

false alarms and allow controlled cancellation. In future work, SOS can be integrated with SMS/WhatsApp alerts or cloud notifications.

5.7 Summary of Results

Table 5.1 summarizes the main observed results across all tests.

Feature	Observed Result	Discussion / Impact
Obstacle Detection	Distance readings updated in real time; safe threshold response observed.	Improves user safety and reduces collision risk; depends on mounting and noise conditions.
Dashboard Monitoring	Live data shown clearly (distance, battery, GPS, pulse).	Supports caregiver awareness and quick decisions; helps validate the system during operation.
Lighting System	Lights successfully activated and improved visibility.	Useful for night/low-light safety; can be enhanced with auto mode in future.
Disconnected Mode	Clear offline feedback; safe stop behavior supported by timeout logic.	Prevents uncontrolled motion if connection is lost; reliability can be improved with fallback links.
SOS Emergency	SOS and STOP SOS actions worked with buzzer/light feedback.	Critical safety function; can be extended with remote alerting and cloud notifications.

Table 5.1: Results and discussion summary for prototype testing.

5.8 Conclusion

This chapter presented practical results from the implemented prototype and verified the key safety and monitoring functions. The results demonstrate that integrating IoT monitoring, obstacle sensing, solar-assisted power, and emergency controls is feasible in a single assistive mobility system. The next chapter concludes the project and presents future development directions.

Chapter 6

CONCLUSIONS AND FUTURE WORK

6.1 Conclusion

This project successfully designed and implemented an IoT-enabled, solar-powered smart wheelchair prototype aimed at improving mobility, safety, and autonomy for individuals with physical disabilities. The system combined hardware components, embedded software, and a web-based user interface into a single integrated solution.

The developed prototype demonstrated the feasibility of using an ESP32-based controller to manage real-time sensing, motor control, wireless communication, and user interaction. Obstacle detection using ultrasonic sensors proved effective in preventing collisions, while the emergency (SOS) mechanism enhanced user safety by providing immediate audible and visual alerts.

The integration of solar power with rechargeable lithium batteries showed the potential for sustainable and off-grid operation, especially in environments with high solar exposure such as Saudi Arabia. The web-based dashboard enabled intuitive control and real-time monitoring without requiring external servers or complex infrastructure.

Overall, the project achieved its main objectives by delivering a functional, low-cost, and scalable assistive mobility prototype that aligns with modern IoT principles and sustainability goals.

6.2 Project Achievements

The key achievements of this project include:

- Successful integration of multiple sensors (ultrasonic, GPS, pulse) for safety and monitoring.
- Reliable motor control using the L298N driver for smooth movement.
- Implementation of a responsive web dashboard for control and monitoring.
- Effective emergency (SOS) feature with clear user feedback.
- Demonstration of solar-assisted power management for extended operation.

These outcomes validate the practicality of combining IoT technologies with renewable energy in assistive healthcare applications.

6.3 Limitations

Despite the successful implementation, the project has several limitations:

- The prototype was tested in controlled and limited environments only.
- Obstacle detection relied on a single ultrasonic sensor, which may not cover all directions.
- Health monitoring was limited to basic pulse indication.
- The solar charging system was evaluated conceptually rather than through long-term field testing.

Addressing these limitations would further improve system robustness and usability.

6.4 Future Work

Future enhancements can significantly extend the capabilities of the smart wheelchair. Potential directions include:

- Integrating additional sensors (e.g., infrared or LiDAR) for 360-degree obstacle detection.

- Adding advanced health monitoring such as blood oxygen level or ECG.
- Implementing cloud-based data storage and analytics for long-term tracking.
- Enhancing navigation with AI-based path planning and autonomous movement.
- Improving the power system with higher-efficiency solar panels and smart energy management algorithms.
- Developing a dedicated mobile application for better user experience.

These future improvements would transform the prototype into a more intelligent, autonomous, and clinically valuable assistive mobility system.

6.5 Final Remarks

The IoT-enabled solar-powered smart wheelchair represents a meaningful step toward accessible and sustainable assistive technology. By combining IoT, renewable energy, and user-centered design, the project contributes to improving quality of life for people with physical disabilities and supports future innovation in smart healthcare and mobility systems.

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